

Game-theoretic evaluation of strategic reasoning in large language models: From complete coverage to compositional complexity

Yu Guo^a, Haochuan Wang^a, Xiachong Feng^{b,*} 

^a Harbin Institute of Technology, Harbin, China

^b The University of Hong Kong, Pok Fu Lam Hong Kong, China

HIGHLIGHTS

- First benchmark covering all 144 canonical games from Robinson-Goforth topology for evaluating AI strategic reasoning.
- Synthetic data generation framework prevents contamination, ensuring reliable evaluation of AI decision-making.
- Hierarchical framework reveals catastrophic performance drops (60% to <20%) as complexity grows, exposing LLM limitations.

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ABSTRACT

Game-theoretic evaluation of strategic reasoning in large language models (LLMs) is crucial for advancing artificial intelligence systems, yet it faces fundamental challenges: incomplete game coverage, data contamination risks, and the inability to assess compositional reasoning complexity. We present TMGBENCH, a benchmark that progresses from complete coverage to compositional complexity through systematic design. For complete coverage, TMGBENCH incorporates all 144 canonical game types from the Robinson-Goforth topology, the first benchmark to achieve exhaustive game-theoretic representation, eliminating the sampling bias that undermines existing evaluations. Each game is instantiated through synthetically generated narrative scenarios, rigorously validated to ensure novelty and prevent data leakage. To address compositional complexity, we introduce a hierarchical framework where these atomic games are programmatically composed into sequential, parallel, and nested structures, creating scalable challenges that systematically probe reasoning depth from simple strategic decisions to complex multi-agent interactions. Our evaluation reveals critical limitations in LLM strategic reasoning across this complete-to-complex spectrum. Even state-of-the-art models fail at basic game-theoretic reasoning, exhibiting logical inconsistencies and superficial Theory-of-Mind understanding. Performance degrades catastrophically as compositional complexity increases: models achieving 60% accuracy on isolated games drop below 20% on compositional structures, exposing fundamental architectural limitations in current AI systems' ability to handle strategic dependencies. These results demonstrate that current LLMs lack the compositional reasoning capabilities required for genuine strategic thinking. TMGBENCH thus provides both comprehensive diagnostic coverage and a scalable complexity framework essential for advancing artificial intelligence toward human-level game-theoretic reasoning and strategic decision-making capabilities.

1. Introduction

The rapid advancement of large language models (LLMs) has reshaped the paradigm of artificial intelligence, achieving breakthroughs across various domains [1–4]. These achievements are largely attributed to LLMs' ability to assimilate vast amounts of knowledge during training,

emerging with the capacity to organize information at a coarse level and link knowledge at a fine-grained level through their internal representations [5]. These core capabilities have driven the success of LLMs in numerous reasoning tasks, including mathematical reasoning [6], commonsense reasoning [7], logical reasoning [8], and strategic reason-

* Corresponding author.

Email address: fengxc@hku.hk (X. Feng).

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0925-2312/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

ing [9,10]. Among these, strategic reasoning has attracted considerable attention due to its multi-agent nature and close association with social intelligence [11,12].

Strategic reasoning refers to the cognitive process of anticipating, planning, and responding to others' actions to achieve specific objectives within competitive or cooperative contexts [13]. Consequently, game scenarios that naturally involve both cooperation and competition, have intuitively become a fertile ground for studying LLMs' strategic reasoning abilities [14]. In particular, researchers have engaged LLMs in game-playing, analyzing their decision-making behaviors and evaluating their strategic intelligence in such scenarios [10]. The Prisoner's Dilemma, as one of the most classic game theory scenarios, has been extensively studied in this context [15]. Additionally, other traditional games such as the Battle of the Sexes [16], the Stag Hunt [17], and the Dictator Game [18] have also drawn significant attention. These studies provide initial insights into the strategic reasoning capabilities of LLMs [19–24].

However, current research has three major limitations, hindering a comprehensive, robust, and sustainable evaluation of LLMs' strategic reasoning capabilities: (1) *Limited coverage of game types*: Most studies focus on a handful of classic games without considering the full diversity of game structures. (2) *Potential risk of game scenario leakage*: Classic game scenarios are likely to be present in the training corpus, raising concerns over data leakage. (3) *Poor extensibility of game forms*: Existing studies primarily focus on a narrow range of game forms, which may no longer suffice to challenge high-performing LLMs such as o3-mini from OpenAI.

To address the above issues, we introduce TMGBENCH, a benchmark that encompasses a comprehensive range of game types, features synthesized game scenarios, and supports scalable and reorganizable game forms. Specifically, *to address the first issue*, we include all 144 game types defined by the Robinson-Goforth topology of 2x2 games [25]. This topology encompasses a variety of game structures based on different numerical payoff matrices, including but not limited to classic games like the Prisoner's Dilemma (Section 3.2). *To address the second issue*, we employ synthetic data generation techniques to create five different story-based games for each classic game. In essence, a story-based game is a contextual framing counterpart of its corresponding classic game, sharing the same structure but differing in context [9]. To ensure high-quality data synthesis, we introduce two additional steps: topic control and human inspection. We first define a set of topics commonly associated with cooperation and competition, such as business and law, to guide the data generation process. Then, to ensure that the synthesized games meet the required game structures and are easily understandable, we conduct rigorous human inspection (Section 3.3). *To address the third issue*, we propose three forms for expanding and organizing games: sequential, parallel, and nested. Using the above constructed games as atomic units, we reorganize them into these complex forms to assess the strategic reasoning of LLMs. The sequential and parallel forms evaluate the model's capacity for sequential and parallel decision-making, respectively, while the nested form explores the LLMs' multi-layered strategic reasoning abilities (Section 3.4).

Based on TMGBENCH,¹ we conduct comprehensive analyses and evaluations of current mainstream LLMs (Section 4), focusing on four core aspects essential for strategic reasoning: (1) *rational reasoning*, the ability to identify optimal strategies through payoff analysis, which forms the foundation of game-theoretic decision-making; (2) *reasoning robustness*, the capacity to maintain consistent strategic choices across different contextual framings of equivalent games [9]; (3) *Theory-of-Mind (ToM) capabilities*, the ability to model opponents' mental states and anticipate their strategies, which is fundamental for higher-order

strategic reasoning [21,26,27]; and (4) *complex game reasoning*, the capacity to handle compositional strategic scenarios. We further validate TMGBENCH through human evaluation studies that establish performance baselines and confirm task quality through systematic comprehension assessment. Our evaluation leads to the following key findings:

- Advanced LLMs (o3-mini, Qwen3, deepseek-reasoner) achieve over 90% accuracy on atomic games, but most models struggle below 60%, revealing that strategic reasoning requires specific capabilities beyond general language understanding.
- LLMs show up to 75% performance reduction on story-based games compared to equivalent classic forms, with high variance across narratives ($c_v \approx 0.5$), indicating reliance on surface patterns rather than abstract game structure understanding.
- First-order ToM prompting benefits some models but not others; second-order ToM provides minimal additional gain, suggesting LLMs lack the recursive reasoning capacity for robust opponent modeling.
- GPT models exhibit systematic asymmetric patterns in 0-task games, with answers correlated to position rather than payoff structure, revealing heuristic-based shortcuts rather than genuine strategic analysis.
- Performance degrades catastrophically with compositional complexity: models drop from 60% on isolated games to below 20% on 10-game compositions, exposing fundamental architectural limitations.

2. Preliminary

2.1. Strategic reasoning

Strategic reasoning [11,13] is a unique and sophisticated form of reasoning that focuses on making optimal decisions in multi-agent environments. It involves carefully selecting strategies by *anticipating the actions of others* and *understanding how one's choices will influence their responses*. What sets strategic reasoning apart from other reasoning paradigms, such as commonsense reasoning, symbolic reasoning, and causal reasoning, is its *dynamic nature* and the *inherent uncertainty of adversarial or cooperative actions*. Unlike these paradigms, strategic reasoning demands a deep comprehension of ever-changing contexts and the ability to make rational, forward-thinking decisions based on the anticipated behaviors of others. Table 1 summarizes these key distinctions.

Real-world applications such as online advertising auctions [28], societal simulation, economic modeling, and gaming [13] frequently require strategic reasoning, as agents must continuously adjust strategies based on others' behaviors. The significance of strategic reasoning for LLMs extends beyond static information retrieval. Equipping LLMs with these capabilities allows them to simulate realistic decision-making, navigate dynamic social or economic systems, and collaborate or compete with other agents, which is crucial for applications such as policy design, automated negotiations, and multi-agent simulations.

Table 1
Characteristics distinguishing strategic reasoning from other reasoning paradigms.

Characteristic	Strategic Reasoning	Other Paradigms
Nature	Dynamic, multi-agent	Typically static, single-agent
Key Challenge	Anticipating others' actions	Understanding patterns, rules, or causality
Uncertainty Source	Adversarial/cooperative actions	Environmental or factual uncertainty
Decision Basis	Anticipated behaviors of others	Known facts, rules, or prior knowledge

¹ Our codes and benchmark are available at <https://www.kaggle.com/datasets/pinkex/tmgbench/>

Table 2
Game-theoretic approaches to evaluating LLM strategic reasoning.

Reference	Game Types	Strategic Reasoning Aspect
Akata et al. [21]	Repeated 2x2 games	Cooperative strategy formation and coordination equilibrium selection
Brookins and DeBacker [14]	Prisoner's Dilemma, Dictator	Strategic cooperation and fairness-driven decision-making
Lorè and Heydar [9]	PD, Stag Hunt, Snowdrift	Abstract strategic reasoning under varied contextual framing
Herr et al. [15]	Stag Hunt, Prisoner's Dilemma	Rational strategy selection under positional and payoff variations
Duan et al. [10]	10 game-theoretic tasks	Strategic reasoning under complete and incomplete information
TMGBENCH (Ours)	All 144 2x2 game types	Comprehensive coverage of all strategic interaction structures

2.2. Strategic reasoning and game theory

Game theory provides the foundational mathematical framework for studying strategic reasoning, which is the process of choosing optimal actions while anticipating the rational responses of other agents [29]. Strategic reasoning is inherently game-theoretic: it requires understanding payoff structures, modeling opponent behavior, and computing equilibrium strategies, capabilities that distinguish it from general language understanding. This theoretical grounding has led recent research to adopt game-theoretic tasks as the standard methodology for evaluating LLM strategic capabilities, as summarized in Table 2.

2.3. Strategic reasoning and theory of mind

Theory of Mind (ToM) is the cognitive ability to attribute mental states (beliefs, intentions, desires) to oneself and others [30]. In strategic reasoning, ToM is essential because anticipating opponents' actions requires understanding their mental states. This recursive modeling distinguishes strategic reasoning from purely computational optimization.

Effective strategic reasoning demands multiple levels of ToM [26]. First-order ToM involves modeling what an opponent will do based on their payoffs. Second-order ToM extends this to consider what the opponent believes you will do. This recursive reasoning is precisely what Nash equilibrium computation requires: each player reasons about the other's best response, which depends on beliefs about one's own strategy.

Recent research explores whether machine ToM exists in LLMs. [27] showed that ToM planning helps LLM agents adapt strategies against adversaries in imperfect information games. [21] found that LLMs perform well in self-interested games but struggle with coordination, and that modeling opponent perspectives improves performance. These findings suggest LLMs may exhibit ToM-like capabilities, but these abilities remain task-dependent and are not yet robust [31].

Given ToM's central role, evaluating whether LLMs possess genuine ToM capabilities is essential. TMGBENCH provides an ideal testbed: its 2x2 games require explicit opponent modeling, and 144 game types ensure comprehensive testing across diverse strategic structures. Through first-order ToM (FoToM) and second-order ToM (SoToM) prompting, we systematically assess whether LLMs engage in genuine recursive opponent modeling or rely on shallow heuristics.

2.4. Benchmark design principles

A comprehensive benchmark for strategic reasoning must systematically cover three fundamental dimensions: (1) *Problem Cardinality*, evaluating both single-game and multi-game reasoning; (2) *Task Dependencies*, ranging from independent games to sequential ordering to interdependent decision-making; and (3) *Representation Modality*, spanning formal payoff matrices to narrative-embedded scenarios. TMGBENCH achieves comprehensive coverage by designing five task

Table 3
The form of a typical 2x2 matrix game.

	Player B	
	B_1	B_2
A_1	(u_A^{11}, u_B^{11})	(u_A^{12}, u_B^{12})
A_2	(u_A^{21}, u_B^{21})	(u_A^{22}, u_B^{22})

types that span all combinations: Atomic (Classic) tests single-game formal reasoning, Atomic (Story) tests single-game narrative comprehension, Sequential tests ordered multi-game reasoning, Parallel tests simultaneous independent reasoning, and Nested tests interdependent multi-game reasoning with forward-looking strategy. Combined with the Robinson-Goforth topology's exhaustive coverage of 144 game types, this design ensures TMGBENCH provides a complete evaluation framework for LLM strategic reasoning. While existing work focuses on only a handful of classic games (e.g., Prisoner's Dilemma, Stag Hunt), TMGBENCH provides the first systematic coverage of all 144 2x2 game types, ensuring comprehensive strategic reasoning evaluation within this well-established paradigm.

2.5. Task formalization

We first formalize the task definition by specifying the game representation, evaluation criteria, and task categories used in TMGBENCH.

Preliminary Definitions. A 2x2 normal-form game $G = (P, S, u)$ consists of two players $P = \{A, B\}$, strategy sets $S_A = \{A_1, A_2\}$ and $S_B = \{B_1, B_2\}$, and payoff functions $u_A, u_B : S_A \times S_B \rightarrow \mathbb{R}$. The payoff structure is represented as a matrix (Table 3), where each cell (i, j) contains the payoff pair $(u_A(A_i, B_j), u_B(A_i, B_j))$ for the corresponding strategy combination.

Nash Equilibrium. A Nash Equilibrium (NE) is a strategy profile $(s_A^*, s_B^*) \in S_A \times S_B$ where no player can unilaterally improve their payoff:

$$u_A(s_A^*, s_B^*) \geq u_A(s_A, s_B^*), \quad \forall s_A \in S_A \quad (1)$$

$$u_B(s_A^*, s_B^*) \geq u_B(s_A^*, s_B), \quad \forall s_B \in S_B \quad (2)$$

The set of all Nash equilibria of the game G is denoted $NE(G) \subseteq S_A \times S_B$. In TMGBENCH, finding $NE(G)$ serves as the evaluation criterion: given a game description, the LLM must identify all Nash equilibria.

Task Types. TMGBENCH organizes evaluation tasks into five types with increasing complexity (Table 4). Each task type is formally defined with explicit input specifications, output requirements, and constraints that govern the evaluation process.

2.6. Notation

Table 5 summarizes the key notation used throughout this paper. The notation is organized into four categories: (1) game-theoretic concepts including game instances, player strategies, payoffs, and Nash equilibria; (2) prompting methods ranging from direct answering to Theory-of-Mind approaches; (3) evaluation metrics for measuring accuracy, inconsistency, and bias; and (4) task categories based on the number of Nash equilibria.

3. TMGBENCH

3.1. Benchmark overview

TMGBENCH is a benchmark designed to evaluate the strategic reasoning capabilities of LLMs in game-theoretic scenarios, illustrated by Fig. 1. It comprehensively covers 144 types of games (see Section 3.2), with each type containing multiple instances (in each instance, there are two players and each player can choose between two strategies, resulting in four possible situations), which can be categorized into

Table 4

Task type categorization in TMGBENCH with formal definitions and constraints.

Task Type	Formal Definition	Input	Output	Constraints & Success Criterion
Atomic (Classic)	Single game reasoning on an explicit payoff matrix; the LLM directly applies Nash equilibrium calculation	Payoff matrix $M \in \mathbb{R}^{2 \times 2}$	$S \subseteq S_A \times S_B$	Direct matrix parsing required; Success: $S = \text{NE}(\mathcal{G})$
Atomic (Story)	Single game reasoning requiring semantic extraction; the LLM must first comprehend the narrative to identify game structure	Narrative text T encoding game $\mathcal{G}$	$S \subseteq S_A \times S_B$	Extract $\mathcal{G}$ from T before reasoning; Success: $S = \text{NE}(\mathcal{G})$
Sequential	Ordered multi game reasoning with temporal constraints; games must be solved in the given sequence	Ordered tuple $(\mathcal{G}_1, \dots, \mathcal{G}_n)$	$(S_1, \dots, S_n)$	Process games in order; no backtracking; Success: $\forall i : S_i = \text{NE}(\mathcal{G}_i)$
Parallel	Simultaneous multi game reasoning; all games are presented at once and must be solved together	Unordered set $\{\mathcal{G}_1, \dots, \mathcal{G}_n\}$	$\{S_1, \dots, S_n\}$	All games given simultaneously; Success: $\forall i : S_i = \text{NE}(\mathcal{G}_i)$
Nested	Two level game with inter game dependency; pre game decisions constrain core game strategy space	Pre game $\mathcal{G}_p$, core game $\mathcal{G}_c$, restriction function R	(S_p, S_c)	$R(S_p)$ restricts strategy space in $\mathcal{G}_c$; Success: Forward looking NE across both levels

Table 5

Notation summary.

Symbol	Definition
<i>Game Theory</i>	
$\mathcal{G} = (P, S, u)$	Game instance (players, strategies, payoffs)
P, S_A, S_B	Players and their strategy sets
u_A^i, u_B^j	Payoffs when A plays A_i , B plays B_j
$S, \text{NE}(\mathcal{G})$	Predicted strategy set; Nash equilibrium set
$\mathcal{G}_p, \mathcal{G}_c$	Pre-game and core-game (nested form)
<i>Prompting Methods</i>	
DA, CoT	Direct Answer; Chain-of-Thought
FoToM, SoToM	First/Second-order Theory-of-Mind
<i>Evaluation Metrics</i>	
PAR, ID, BD, SD	Perfect Accuracy Rate; Inconsistency/Bias/Significance Degree
$\text{Freq}_{i,j,o}$	Frequency of outcome o at grid (i, j)
<i>Task Categories</i>	
n -task	Task with n Nash equilibria ($n \in \{0, 1, 2\}$)

Section 3.3). Furthermore, each game in TMGBENCH can be treated as an atomic unit, and multiple atomic games can be structured into a more complex task with parallel, sequential, or nested forms (see Section 3.4). These complex scenarios effectively facilitate the evaluation of advanced LLMs' abilities in parallel, sequential, and multi-layered decision-making. To precisely evaluate the reasoning abilities of LLMs, we use their performance in inferring the optimal strategy combination, i.e., the *Nash equilibrium*, as the evaluation criterion. Additionally, the designed evaluation metrics provide a fine-grained assessment of the robustness and self-consistency of LLMs' strategic reasoning abilities (see Section 3.5).

3.2. Game topology

Although previous research has explored the reasoning abilities of LLMs within game theory, existing studies have primarily focused on a few well-known games, such as the Prisoner's Dilemma and Stag Hunt [14,20,32]. This narrow focus results in incomplete evaluations and highlights the urgent need for a broader variety of games to conduct a systematic assessment.

To address this gap, we first establish the fundamental unit of our analysis: the 2×2 normal-form game. As formalized in Section 2.5, a 2×2 game $\mathcal{G}$ involves two players choosing between two strategies, with payoffs specified in a matrix (Table 3).

While a single game matrix defines one interaction, a systematic evaluation requires covering all possible strategic structures. For this, we adopt the Robinson-Goforth topology of 2×2 games, which classifies all 144 distinct strategic scenarios [25]. This framework is organized as a 12×12 grid, where well-known symmetric games are situated on the counter-diagonal (including famous games shown in Fig. A.11(a) in Appendix A), and asymmetric games are paired with a "sister game" across the diagonal (see Fig. A.11 in Appendix A for the complete topology visualization with a detailed grid structure in Fig. A.11(b)).

The topology's true power for evaluation lies in its distribution of solutions. The 144 games are categorized into three distinct groups based on their number of Nash equilibria, stable outcomes where neither player has an incentive to unilaterally change their strategy. There are 108 games with a single Nash equilibrium (where at least one player has a dominant strategy), 18 with two, and 18 with none (where at least one player always has an incentive to deviate). This distribution of solution types is symmetric across the counter-diagonal, presenting a diverse and challenging landscape for testing LLM reasoning (see Fig. A.12 in Appendix A for a visual representation of this distribution). We categorize all 144 games, using the numerical payoffs from the original topology, as the *classic game setting* for our benchmark.

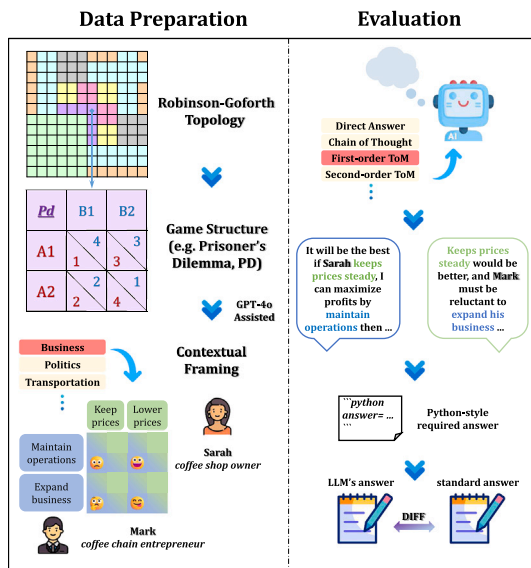


Fig. 1. Conceptual overview of TMGBENCH with three key components: Robinson-Goforth topology, game structure, and contextual framing.

classic and story-based settings. Notably, the story-based instances are produced using synthetic data generation techniques and are grounded in real-life themes, effectively mitigating the issue of data leakage (see

3.3. Contextual framing

Relying on the Robinson-Goforth topology, we can systematically construct all types of classic setting tasks. However, this alone is insufficient, as games often take place in diverse real-life contexts, involving different topics, types of participants and their preferences. Such contextual framing of games introduces new challenges for LLMs [9].

To further explore LLMs' strategic reasoning capabilities in real-world scenarios, we use classic games as seed data and employ synthetic data generation techniques, leveraging GPT-4o to construct story-based games. Specifically, in story-based games, we replace the pure game information of classic games with real-life scenarios, covering topics such as business, law and transportation. Additionally, the two players are substituted with characters representing broader semantics (e.g., people, animals, organizations, and even nations), and the payoff values are transformed from pure numbers into specific states or rewards relevant to the characters. For each classic game, we generate 5 corresponding story-based games. To ensure the generation of a high-quality and diverse dataset, the pipeline is organized into three carefully designed stages:

Classic Game Construction. Our process begins by converting the 144 abstract payoff matrices from the topology into classic games. Through context engineering [33], we craft a specific description for each game type. The example below, for instance, mirrors the structure of the Prisoner's Dilemma. These 144 classic games provide the structural blueprint for the next stage: generating a wider variety of rich, story-based games.

An example of a classic game with a 2x2 payoff matrix structure similar to the Prisoner's Dilemma is provided in [Appendix B.1](#).

Story-based Game Generation. While classic games provide robust mathematical formalisms, real-world strategic interactions are seldom context-free; they are deeply embedded in rich social narratives. To bridge this gap between abstract theory and applied reasoning, we developed story-based games designed to simulate this complexity with narrative depth and situational details.

To generate these scenarios systematically, we employed a synthetic data generation pipeline guided by meticulously crafted prompts. These prompts imposed specific structural constraints to ensure each narrative accurately reflected its underlying game-theoretic structure. To ground these narratives in plausible contexts, we curated a set of 20 diverse thematic categories drawn from real-world situations, such as business negotiations or family disputes. Following this procedure, we generated 5 unique story-based counterparts for each of the 144 classic games. The prompt structure governing this synthesis, with variable placeholders indicated in red, is detailed below.

The detailed prompt template used for generating story-based games with GPT-4o is provided in [Appendix B.2](#). This prompt imposes specific structural constraints to ensure each narrative accurately reflects its underlying game-theoretic structure while grounding it in one of 20 diverse thematic categories.

The resulting story-based games are designed to be more than just re-skinned matrices; they are complex, narrative-rich scenarios that better reflect the ambiguity and contextual nature of real-world strategic decisions. This narrative layer introduces a significant challenge, as models must first parse and comprehend the story to extract the underlying game structure, a task that is fundamentally more demanding than solving a pre-defined matrix. The example below illustrates this leap in complexity and realism, serving as a narrative counterpart to the classic Prisoner's Dilemma game.

An example of a generated story-based game featuring Sarah's coffee shop versus Mark's coffee chain, which follows the same game structure as the Prisoner's Dilemma, is provided in [Appendix B.3](#). This example illustrates the leap in complexity and realism from abstract payoff matrices to narrative-rich scenarios.

Quality Verification. To guarantee the coherence and internal consistency of each narrative, we employ a rigorous two-stage human

validation process. In the first stage, GPT-4o generates a draft scenario, which is then subjected to meticulous human review to identify any logical flaws. If a draft fails this inspection, it enters an iterative refinement cycle where GPT-4o is prompted to self-diagnose and correct the problematic sections. This process continues until the narrative meets our stringent quality standards; scenarios that repeatedly fail are discarded entirely to maintain the integrity of the dataset. In the second stage, we conducted a formal comprehension validation study to ensure the finalized scenarios are understandable and preserve game-theoretic structure. Three expert annotators (graduate students with a game theory background) evaluated all 144 story-based scenarios across five criteria: narrative clarity, role clarity, option clarity, payoff clarity, and structure preservation, each on a 1–5 Likert scale. Results demonstrate high overall quality (mean = 4.34/5.0, SD = 0.62) with substantial inter-annotator agreement (Fleiss' κ = 0.78). Role clarity scored highest (4.52), confirming that player identities are well-defined, and structure preservation scored 4.43, indicating accurate representation of game-theoretic relationships. Importantly, 98.6% of scenarios (142/144) received overall ratings ≥ 4.25 , confirming consistent quality across the benchmark.

Benchmark Diversity. A key feature of our story-based tasks is their designed diversity in both thematic context and narrative length. To ensure contextual richness, we established 20 distinct topics from everyday life, such as Business, Ecology, and Politics. The distribution of these topics is intentionally varied, with Business (11.1%) as the most prominent category and the remainder spread evenly to ensure broad coverage. In parallel, we controlled for narrative length, with most scenarios ranging from 200 to 450 words and over 90% concentrated between 250 and 400 words. This systematic variation in both what the stories are about and how long they are creates a robust and comprehensive testbed for evaluating LLM reasoning (see [Fig. A.13](#) in [Appendix A](#), which shows the topic distribution in [Fig. A.13\(a\)](#) and narrative length patterns in [Fig. A.13\(b\)](#)).

3.4. Complex forms

The 2x2 games in the topology represent a highly condensed game structure. However, in real life, we often encounter more complex game forms, such as making continuous decisions, making multiple decisions simultaneously, or considering the impact of one decision on another.

To evaluate LLMs' strategic reasoning abilities with more constraints, we treat the aforementioned individual games as atomic games and expand them in three forms: sequential, parallel, and nested. The organization of these forms is illustrated in [Fig. 2](#). Specifically, in the *sequential form*, we randomly sample multiple games from the story-based games, requiring the LLM to make decisions sequentially. Only if the LLM provides correct answers for all games is it considered to have made correct decisions. In the *parallel form*, the LLM is given multiple randomly sampled games and must make decisions simultaneously. Similarly, the LLM is deemed to have made correct decisions only if it solves all games correctly. In the *nested form*, we randomly sample two games, designated as the pre-game and the core-game, where the core-game holds greater importance. The decisions made by the LLM in the pre-game affect the strategy space in the core-game. Thus, the LLM is judged to have made correct decisions only if it demonstrates forward-looking reasoning by choosing a sub-optimal solution in the pre-game to achieve the optimal solution in the core-game. This design mirrors *extensive-form games* [34] where sequential decisions create outcome-contingent strategy constraints. The restriction rule is explicitly stated in the prompt as a conditional constraint, ensuring logical consistency with standard game-theoretic multi-stage dynamics. We demonstrate a template to generate a nested form game in [Appendix B.4](#).

After a nested form game is generated through our template, we still need to check if the Nash equilibria of the pre-game change after the restriction from the core game. If the set of Nash equilibria does change,

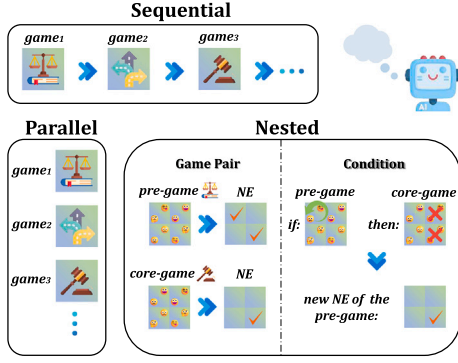


Fig. 2. Three complex game forms in TMGBENCH: sequential (games in sequence), parallel (simultaneous games), and nested (linked pre-game and core-game).

then we use this as a piece of data to evaluate LLMs, observing whether they can detect such a violation of the original NEs' structure.

Theoretically, using these atomic games, we can expand the framework to generate infinitely many increasingly complex game forms, thereby providing a continuous benchmark for evaluating the performance of more advanced LLMs. Fig. 3 illustrates how we build sequential, parallel, and nested games from atomic games.

3.5. Evaluation metrics

As explained in Section 3.2, our benchmark is perfectly suitable for display in a 12x12 square table, with each grid representing one of the 144 equivalence classes. In the evaluation process we conduct *repetitive* tests at every data point of each equivalence class. Each test starts with the input of the setting (classic/story-based) and the question, and ends with the LLM's response containing a list of choices corresponding to multiple choices or no choice (when the given list is empty).

Evaluation Process. Algorithm 1 formalizes our evaluation procedure for assessing LLM strategic reasoning capabilities.

Inconsistency Heat Map. According to the conclusions of the Robinson-Goforth topology [25], we convert the standard answer of each equivalence class into a heat map named the *standard heat map*, with the colored quarter-grid being the choice in the standard answer. Similarly, as for the practical result provided by LLMs, we set the value of $\text{Freq}_{i,j,o}$ as the color depth of each quarter grid, which builds up the *practical heat map*. Naturally, we subtract the standard heat map from the practical heat map in an element-wise manner to get the *inconsistency heat map*, which is a standardised tool for our evaluation as shown in Fig. 4.

Inconsistency Degree. In order to display the quantified performance of LLMs, we extract the inconsistency degree from a map, which helps reveal the gap between LLMs' responses and the standard answer, and it is defined as: $\text{ID} = \frac{1}{144} \sum_{i=1}^{12} \sum_{j=1}^{12} \frac{1}{4} \sum_{o=1}^4 \Delta \text{Freq}_{i,j,o}^2$, where $\Delta \text{Freq}_{i,j,o}$ indicates the difference (between the LLM's answer and the standard answer) in the frequency of the o -th choice at the i -th row, j -th column.

Bias Degree. Owing to the symmetric property of the topology framework of 2x2 matrix games, the distribution of answers over the heat map has axial symmetry along the counter-diagonal as shown in Fig. 5. Motivated by this elegant property, we set up another metric to evaluate the bias degree of LLMs' answers, as we expect more robust LLMs to display lower degrees of bias. The bias degree reflects the stability and symmetry of LLMs' strategy, and it is defined as: $\text{BD} = \frac{1}{144} \sum_{i=1}^{12} \sum_{j=1}^{12} \frac{1}{4} \sum_{o=1}^4 (\text{Freq}_{i,j,o} - \text{Freq}_{j,i,\text{ref}_o})^2$, where the meaning of ref_o is the index of choice o 's counterpart considering the reflection operation along the counter-diagonal, and we have the mapping relation: $\{1, 2, 3, 4\} \mapsto \{4, 2, 3, 1\}$. (e.g., $\text{ref}_1 = 4$ means that the reflection counterpart of choice 1 is choice 4, and vice versa)

Perfect Accuracy Rate. In addition to the metrics mentioned above, we also set up a more rigorous metric named perfect accuracy rate, which ignores the partially correct answers and only considers perfectly correct answers in each test, and it is defined as: $\text{PAR} = \frac{1}{144} \sum_{i=1}^{12} \sum_{j=1}^{12} \frac{1}{T} \sum_{t=1}^T \mathbb{I}\{\text{rsp}_{t,i,j} = \text{std}_{i,j}\}$, which means that we count only if the response perfectly matches the standard answer, where T represents the number of times we invoke a LLM to respond to a certain game task. Notably, the 144 game types in our topology exhibit varying equilibrium structures: 108 games have exactly one Nash equilibrium, 18 games have two Nash equilibria, and 18 games have no pure-strategy Nash equilibrium. For *no-NE games* (18 games), the standard answer is an empty set ($\text{NE}(G) = \emptyset$); the model must correctly identify that no pure-strategy equilibrium exists. For *multiple-NE games* (18 games), the model must identify the *complete set* of all equilibria. Identifying only one valid NE is insufficient. The PAR criterion requires *exact set equality*: the predicted set S must match $\text{NE}(G)$ precisely, neither missing any equilibrium nor including spurious ones. This strict criterion ensures that high PAR scores reflect genuine game-theoretic understanding rather than partial pattern matching.

Metrics with Subscript. As a matter of fact, within the topology, different equivalence classes have different numbers of Nash equilibria (ranging from $\{0, 1, 2\}$), leading to a discrepancy in reasoning difficulty, we propose metrics with subscripts that represent different types of equivalence groups (we refer to them to 0-task, 1-task, and 2-task respectively), which we refer as as sub-metrics. Therefore we have $\text{ID}_n, \text{BD}_n, \text{PAR}_n$ ($n = 0, 1, 2$) which mean the inconsistency degree, the bias degree, and the perfect accuracy rate across all equivalence classes that have n equilibria.

Detailed worked examples demonstrating step-by-step calculations for all evaluation metrics, including Inconsistency Degree (ID), Bias Degree (BD) and Perfect Accuracy Rate (PAR) are provided in Appendix E.

4. Analysis

4.1. Overview of LLMs' performance

Overall, we select several SOTA models according to the Open LLM Leaderboard [35] and conduct extensive experiments on TMGBENCH. These models include GPT (o3-mini, gpt-4o, gpt-4o-mini, gpt-3.5-turbo), Claude (claude-3-sonnet, claude-3-haiku), Llama (Llama-3.1-8B, Llama-3.1-70B), Gemma (gemma-3-27b-it), Qwen (Qwen3-32B, Qwen2-72B) and Deepseek (deepseek-reasoner). We perform 4 independent tests on each data point, covering both the classic setting and the story-based setting (thus we conduct 2880 tests to evaluate a certain model). During the evaluation, we set the generation temperature to 0, ensuring the lowest degree of uncertainty and enhancing the faithfulness of the evaluation.

Games in TMGBENCH are not easy for most LLMs. We initially adopt two basic prompting methods: Direct Answer (DA) prompting and Chain-of-Thought (CoT, [36]) prompting, which represent shallower, faster thinking patterns and deeper, slower thinking patterns, respectively. All prompting methods used in our experiments are detailed in Appendix C. The Direct Answer (DA) prompting method is detailed in Appendix C.1. The Chain-of-Thought (CoT) prompting method is detailed in Appendix C.2. All prompting methods require structured Python-style output for automated evaluation; a sensitivity analysis confirming that this format does not interfere with reasoning is provided in Appendix G.

As seen from Table 6, o3-mini, Qwen3-32B, and deepseek-reasoner are more capable compared to other models, with a high overall accuracy rate (over 90%) and low inconsistency and low bias scores (around 1%). Specifically, as shown in Fig. 6 formed by 9 sub-metrics, Qwen3-32B and deepseek-reasoner almost surpass any other models in every aspect. Moreover, comparing the performance of employing DA

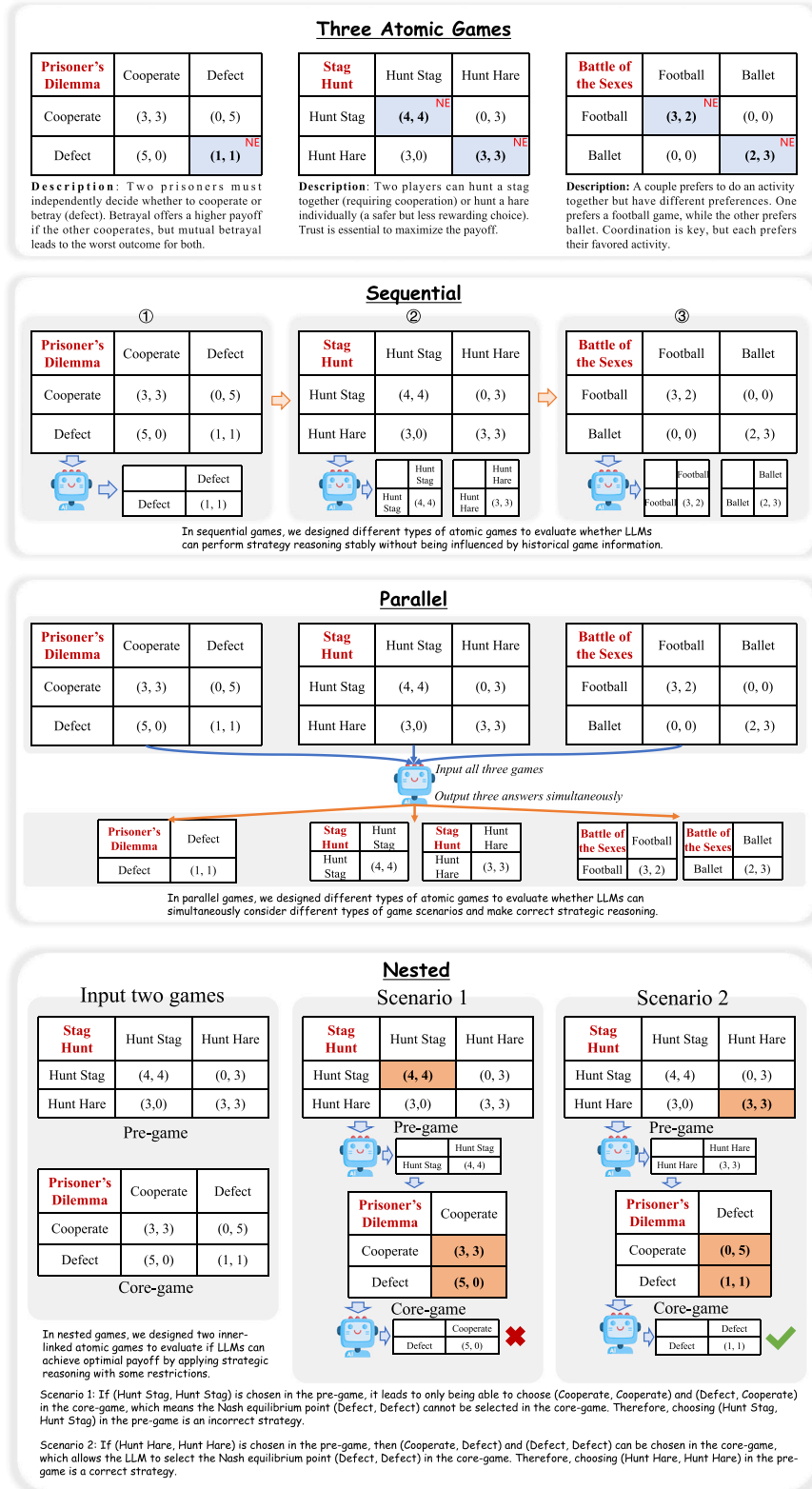


Fig. 3. An example of how we build sequential, parallel, and nested game from some of the atomic games in TMGBENCH.

prompting and CoT prompting, we find that CoT prompting almost provides comprehensive improvement. Despite the excellent performance of Qwen3-32B and deepseek-reasoner, other models often do not exhibit robust performance across all 3 different types of tasks. This indicates

that even classic setting tasks from TMGBENCH are challenging for most LLMs.

The sharp capability gap (most models fall below 60% accuracy) reveals that strategic reasoning is not simply a byproduct of general

Algorithm 1 TMGBENCH Evaluation Procedure.**Require:** Game instance $\mathcal{G}$, LLM $\mathcal{M}$, Prompting method P **Ensure:** Correctness indicator, Predicted set S , Ground truth $NE(\mathcal{G})$

- 1: prompt $\leftarrow \text{CONSTRUCTPROMPT}(\mathcal{G}, P)$
- 2: response $\leftarrow \mathcal{M}(\text{prompt})$
- 3: $S \leftarrow \text{PARSERESPONSE}(\text{response})$ {Extract strategy set $S \subseteq S_A \times S_B$ }
- 4: correct $\leftarrow (S = NE(\mathcal{G}))$ {Check exact match}
- 5: **return** correct, S , $NE(\mathcal{G})$

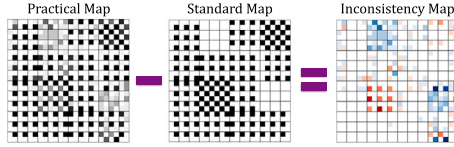


Fig. 4. Inconsistency heat map showing differences between LLM responses and standard answers. Blue indicates positive difference, red indicates negative difference.

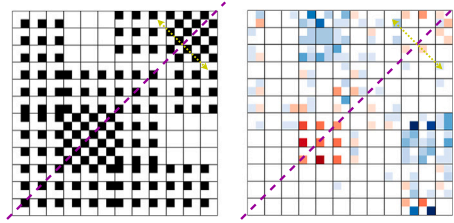


Fig. 5. Axisymmetry comparison: standard heat map (left) shows perfect counter-diagonal symmetry, while LLM responses (right) exhibit quasi-axisymmetry with position-specific discrepancies.

language ability. The models that succeed share explicit reasoning architectures designed for multi-step logical analysis, suggesting that strategic reasoning demands specific computational capabilities beyond standard natural language tasks.

Finding 1: Basic Strategic Reasoning Remains Challenging

While advanced reasoning models (o3-mini, Qwen3-32B, deepseek-reasoner) achieve >90% accuracy on atomic games, most LLMs struggle below 60%. This capability gap suggests strategic reasoning requires specific skills, namely systematic payoff analysis and equilibrium identification, beyond general language understanding.

LLMs' performance is vulnerable across various narratives. At the theoretical level, we consider classic setting tasks and story-based tasks to be fundamentally the same problems within the domain of game theory. However, this conclusion does not appear to be transferable to LLMs at the practical level. For LLMs, the complexity and nuance of story-based tasks introduce unique challenges, where LLMs are required to be robust in understanding and reasoning concurrently.

In Fig. 7, we compare the performance of LLMs using CoT prompting, which is more robust according to previous analysis. The figure reveals the vulnerable performance of LLMs on tasks in a story-based setting (corresponding to various narratives), marked by two primary characteristics: (1) Some models like o3-mini, gpt-4o, claude-3-5-sonnet and gemma-3-7b-it, exhibit significant performance degradation. Notably, o3-mini, gpt-4o and gemma-3-7b-it demonstrate a broad under-performance, while claude-3-5-sonnet shows the largest drop in 0-task, with its $S\text{-PAR}_0$ metric reduced to less than one-fourth of $C\text{-PAR}_0$, and its $S\text{-ID}_0$ metric exceeding four times that of $C\text{-ID}_0$. (2) The performance of certain localities exhibits significant fluctuations. A particularly notable

degradation occurs in the PAR scores for 0-task and 2-task scenarios handled by claude-3-5-sonnet, where the coefficients of variation c_v (defined as $c_v = \frac{\sigma}{\mu}$, with σ representing the standard deviation and μ the mean) approach 0.5. These eminent values of c_v suggest a lack of robustness in performance across different narratives.

This sensitivity to surface presentation while game structure remains constant indicates that LLMs do not learn abstract representations of game-theoretic relationships. Instead, they appear to rely on pattern matching with familiar scenarios from training data. When narrative framing changes, these learned associations break down, revealing the absence of genuine structural understanding.

Finding 2: Performance is Context-Dependent but Not Structure-Aware

LLMs show up to 75% performance reduction when equivalent games appear in narrative form, with high variance across stories. Models rely on surface patterns rather than abstract game structure understanding, which constitutes a fundamental barrier to robust strategic reasoning.

Human Performance Baseline. To contextualize LLM performance and validate task difficulty, we conducted a human baseline study. Five graduate students with a game theory background (economics, computer science, mathematics) completed 72 tasks: 36 classic and 36 matched story-based scenarios (12 each from 0-task, 1-task, 2-task categories), yielding 180 responses per setting. Human participants achieved 86.7% accuracy on classic tasks (156/180 correct) and 82.2% on story-based tasks (148/180 correct), a modest 5.2% relative reduction (Table 7). This contrasts sharply with LLMs: average models dropped from 60.7% to 46.5% (23.4% relative reduction), while even top models dropped from 94.5% to 78.7% (16.7% reduction). The consistency of human performance across representations validates two key points: (1) story-based tasks preserve the underlying game structure, as humans can still solve them effectively; and (2) the larger LLM performance drop reflects genuine limitations in handling narrative-embedded strategic reasoning, not task design artifacts. Notably, humans required significantly more time for story-based tasks (73s vs. 42s per task), indicating additional cognitive load from narrative comprehension, yet maintained accuracy, a capability current LLMs lack.

4.2. Findings of LLMs' behaviors

LLMs demonstrate first/second-order ToM abilities. In all equivalence classes of tasks, 1-tasks have the lowest reasoning difficulty since at least one player has a dominant strategy, which means the player can make an unconditionally optimal decision regardless of the counterpart's choice. In such cases, once a player (denoted as A) can make an unconditionally optimal decision, their counterpart (B) can, using first-order Theory-of-Mind (ToM), easily determine the best response for themselves (B).

This insight motivated us to apply FoToM prompting to LLMs, representing the First-order Theory-of-Mind thinking, to aid in solving these tasks. As seen in Table 8, gpt-4o shows improvement in both 0-tasks and 1-tasks when utilizing FoToM. Model claude-3-5-sonnet improves on 1-tasks and 2-tasks, and gpt-4o-mini displays a significant surge in performance on 1-tasks and so does Llama-3.1-70B on 2-tasks. However, no LLM achieves overall improvement across all task categories by merely using first-order ToM, and 0-tasks appear to be the most challenging for LLMs to solve. The First-order Theory-of-Mind (FoToM) prompting method is detailed in Appendix C.3.

Furthermore, we wondered if LLMs that display some ability to use first-order ToM could also be capable of second-order ToM. According to [37], higher-order ToMs are generally more difficult to master than first-order ToM. Thus we selected only advanced models that demonstrated proficiency in first-order ToM to attempt solving specific tasks using Second-order Theory-of-Mind (SoToM) prompting. As seen in Table 8, gpt-4o, gpt-4o-mini and claude-3-5-sonnet show consistent performance when applying second-order ToM to tasks they

Table 6

Overall statistics of LLMs' performance on classic setting tasks. The up arrow($\uparrow$) means the a larger value indicates better performance, while that down arrow($\downarrow$) means that a smaller value indicates better performance. All values are expressed as percentages.

Family	Model	Metric / Prompting					
		PAR($\uparrow$)		ID($\downarrow$)		BD($\downarrow$)	
		DA	CoT	DA	CoT	DA	CoT
GPT	o3-mini	93.40	93.23	1.91	2.17	3.82	4.34
	gpt-4o	52.08	80.38	16.81	3.78	28.49	7.79
	gpt-4o-mini	14.93	74.02	27.15	4.38	48.59	8.29
	gpt-3.5-turbo	30.21	34.38	27.64	17.87	50.15	30.19
Claude	claude-3-5-sonnet	59.38	79.69	14.79	7.13	27.76	14.34
	claude-3-haiku	24.31	40.28	39.58	25.17	72.22	44.10
Llama	Llama-3.1-70B	13.02	54.29	36.15	15.32	40.71	26.63
	Llama-3.1-8B	18.75	22.63	38.49	31.19	81.32	47.64
Qwen	Qwen3-32B	99.13	98.78	0.09	0.31	0.17	0.63
	Qwen2-72B	43.06	46.21	26.30	19.94	35.59	29.29
Google	gemma-3-27b-it	20.83	86.11	35.15	3.31	63.35	6.55
Deepseek	deepseek-reasoner	98.78	97.74	0.09	0.29	0.17	0.59

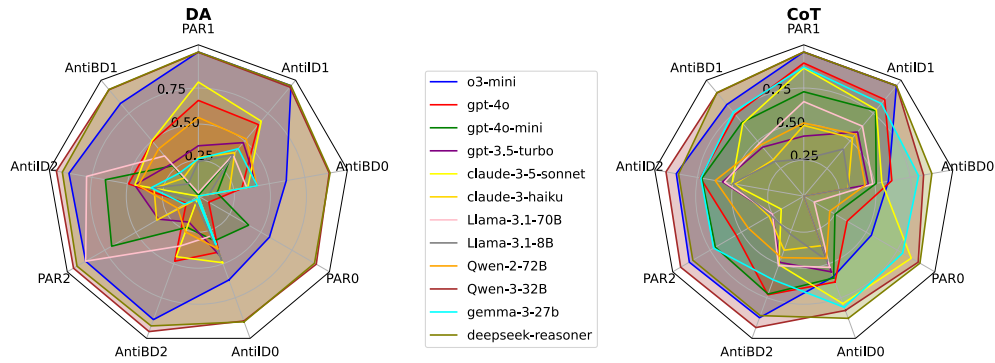


Fig. 6. Radar charts of the 9 sub-metrics of LLMs' performance, comparing the DA prompting (left side) and the CoT prompting (right side). AntiID and AntiBD are derived from ID and BD while higher values indicate better performances (in order to consistent with PAR) ($\text{AntiBD} = 1 - \sqrt{\text{BD}}$, $\text{AntiID} = 1 - \sqrt{\text{ID}}$).

are already capable of solving better with first-order ToM. However, the improvements from using SoToM generally do not exceed those achieved with first-order ToM. In addition, Llama-3.1-70B's underperformance with SoToM suggests that possessing first-order ToM capabilities does not necessarily imply proficiency with second-order ToM. The Second-order Theory-of-Mind (SoToM) prompting method is detailed in Appendix C.4.

The inconsistent benefits of ToM prompting reveal a fundamental limitation: strategic reasoning requires recursive modeling of opponents' beliefs about one's own strategy. Models that benefit from FoToM can simulate one level of opponent reasoning, but the minimal gains from SoToM suggest they cannot recursively apply this process, as they lack the architectural capacity for genuine recursive computation over mental states.

Finding 3: Theory-of-Mind Capabilities Are Limited and Inconsistent

First-order ToM prompting benefits some models (gpt-4o, claude-3-5-sonnet) but not others, while second-order ToM provides minimal additional gain. LLMs lack robust opponent modeling, which is a core requirement for strategic interaction that demands recursive reasoning about others' mental states.

Certain behavioural patterns contribute to poor performance.

Based on the analysis from the previous sections, it is encouraging to note that most LLMs demonstrate the best performance when solving 1-task scenarios, regardless of the prompting used (CoT, FoToM, or SoToM). However, some models, specifically gpt-4o and gpt-4o-mini, experience a significant decline in performance when addressing other types of tasks, and it is particularly noteworthy that they perform the

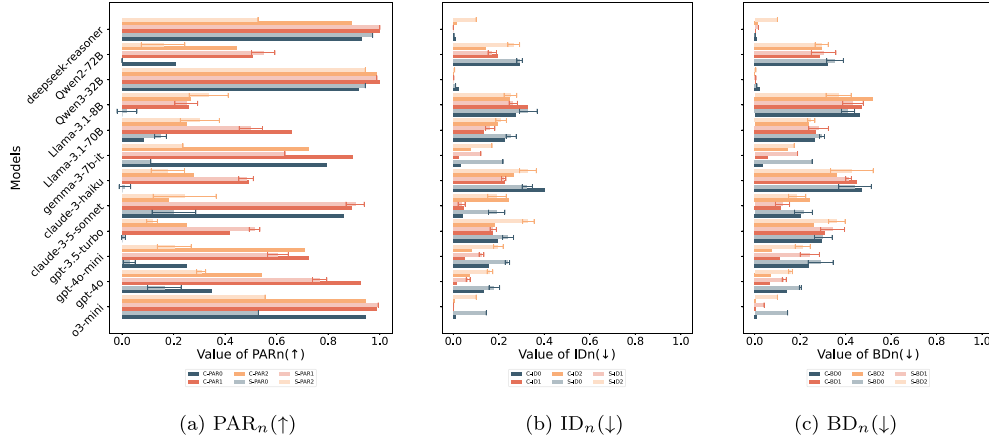


Fig. 7. Comparison of LLMs' performance under classic ('C', opaque) and story-based ('S', semi-opaque with error bars) settings. Bar length represents metric values; error bars show standard deviation across 5 story-based data points.

Table 7

Human vs. LLM performance comparison across settings and task types. Human accuracy remains stable across representations, while LLMs show significant degradation on story-based tasks. Human results are based on 5 participants $\times$ 12 tasks per category = 60 responses per cell.

Agent	0-task	1-task	2-task	Overall
<i>Classic Setting</i>				
Human	78.3%	95.0%	86.7%	86.7%
Best LLM (o3-mini)	91.2%	97.8%	94.5%	94.5%
Average LLM	52.3%	71.5%	58.2%	60.7%
<i>Story-based Setting</i>				
Human	73.3%	91.7%	81.7%	82.2%
Best LLM (o3-mini)	68.5%	89.2%	78.3%	78.7%
Average LLM	38.7%	56.8%	44.1%	46.5%

worst on 0-tasks out of all types. Surprisingly, as illustrated in Fig. 8, these models display a similar answering pattern that appears non-coincidental. Within the topological framework, there are two square areas representing 0-tasks (enclosed in yellow boxes and green boxes), which should theoretically be symmetric across the counter-diagonal. The standard heat map of these two areas is entirely blank, reflecting no existing equilibrium, so the two areas of the inconsistency heat maps just reflect the distribution of LLMs' practical responses. Thus it is evident that the models exhibit a consistent pattern when addressing 0-tasks. In yellow-box areas, their answers tend to emphasize the upper-right and lower-left quarter-grids, whereas in green-box areas, their answers tend to emphasize the upper-left and lower-right quarter-grids. This position-dependent pattern (rather than random errors) suggests systematic bias in the models' decision-making process.

To mitigate the possible effects of the position bias in the *FoToM* prompting and *SoToM* prompting, we design the *reFoToM* prompting (see Appendix C.5) and the *reSoToM* prompting (see Appendix C.6), which swap the order of the players as it occurs in the *FoToM* prompting and the *SoToM* prompting respectively.

We show in Fig. 10 that GPT series models still display a similar pattern when using *reFoToM* and *reSoToM* prompting. Yellow-box areas and green-box areas display an asymmetric inconsistency pattern. The Reversed First-order Theory-of-Mind (*reFoToM*) prompting method is detailed in Appendix C.5. The Reversed Second-order Theory-of-Mind (*reSoToM*) prompting method is detailed in Appendix C.6.

In order to further quantify how significant the results display such a pattern, we also propose a metric named significance degree which is confined to $[0, 1]$, defined as:

$$SD = \frac{1}{18} \sum_{i,j} \mathbb{I}\{\#NE(i,j) = 0\} \cdot \frac{1}{4} \times (\text{Freq}_{i,j,1} + \text{Freq}_{i,j,4} - \text{Freq}_{i,j,2} - \text{Freq}_{i,j,3}) \times S(i,j) \quad (3)$$

where $\#NE(i,j)$ represents the number of Nash equilibria of the tasks of grid (i,j) , and $S(i,j)$ is determined by the area that the grid (i,j) belongs to, having a value of 1 if the grid is in the green area and a value of -1 if the grid is in the yellow area.

We present the statistical results of LLMs' performance in Table 9, which show that the SD values for using *ReFoToM* are similar to those for *FoToM*, and the values for *ReSoToM* are close to those for *SoToM*. Additionally, the results indicate that employing ToM helps gpt-4o reduce the asymmetric inconsistency pattern, while it conversely makes gpt-4o-mini more 'stubborn' in maintaining this pattern. Furthermore, higher-order ToM appears to have a stronger effect than first-order ToM.

The systematic nature of these patterns is revealing. If models were performing genuine payoff analysis, their errors would be randomly distributed across equivalent game positions. Instead, the consistent correlation between answer patterns and game positions suggests that models use positional heuristics (likely learned associations between answer positions and success in training data) as shortcuts for strategic decision-making.

Finding 4: Systematic Biases Reveal Heuristic-Based Decision Making

GPT models exhibit consistent asymmetric patterns in 0-task games, with answers correlated to game position rather than payoff structure. This indicates reliance on positional shortcuts rather than genuine strategic analysis.

To verify that these asymmetric patterns originate from model reasoning rather than potential bias in story generation (e.g., Player A being framed as the protagonist), we conducted a role-swap sensitivity analysis. The results confirm that swapping character assignments has minimal impact on performance, indicating that the underlying game structure (not narrative role assignment) determines model behavior (see Appendix H).

Complex forms bring more challenging tasks. To verify that TMGBENCH can be extended to harder tasks which may better align with complicated scenarios from reality, we run the test on the three complex forms mentioned in Section 3.4, to assess the performance of four top-performing LLMs (o3-mini, gpt-4o, Qwen2-72B, and deepseek-reasoner) in complex strategic reasoning. Full experimental configurations for reproducing these results are provided in Appendix F. We set up the

Table 8

Performance of LLMs using different ToM compared to CoT. Text in red color indicates that the performance gets better and text in blue color indicates that the performance declines (both compared to CoT). Bold text represents the best performance across the three prompting methods. Grey areas indicate that an LLM is proficient in using some kind(s) of ToM. All values are expressed as percentages.

Model	Prompting	0-Task			1-Task			2-Task		
		PAR ₀ (↑)	ID ₀ (↓)	BD ₀ (↓)	PAR ₁ (↑)	ID ₁ (↓)	BD ₁ (↓)	PAR ₂ (↑)	ID ₂ (↓)	BD ₂ (↓)
gpt-4o	CoT	34.72	13.37	14.41	92.36	1.58	6.76	54.17	7.38	7.38
	FoToM	43.06	9.46	9.81	95.14	0.72	4.14	50.00	8.94	8.59
	SoToM	31.94	9.81	10.68	91.67	1.45	6.00	52.78	7.99	8.16
gpt-4o-mini	CoT	25.00	15.62	23.94	72.45	5.08	11.09	70.83	7.97	7.69
	FoToM	25.00	19.53	19.53	99.54	0.03	5.08	47.22	10.59	10.59
	SoToM	18.06	26.56	26.22	98.84	0.19	5.38	68.06	5.38	5.38
claude-3-5-sonnet	CoT	86.11	4.25	20.23	88.89	4.72	11.68	18.06	24.48	24.48
	FoToM	68.06	7.73	16.06	92.13	2.56	7.74	47.22	15.10	15.10
	SoToM	47.22	21.35	28.99	90.05	4.05	14.38	33.33	14.93	14.93
Llama-3.1-70B	CoT	8.33	22.47	26.43	65.59	13.43	27.16	25.00	19.53	23.70
	FoToM	2.78	30.82	35.59	49.54	18.68	27.49	69.44	6.08	22.74
	SoToM	23.61	21.27	28.73	60.42	14.09	23.70	12.50	24.05	25.26

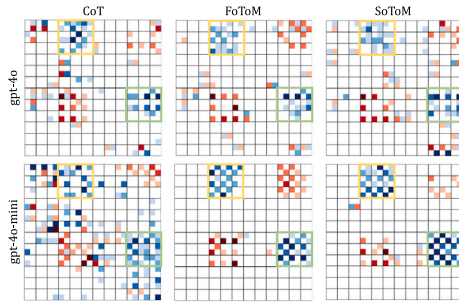


Fig. 8. Inconsistency heat map of gpt-4o and gpt-4o-mini using different prompting methods. The yellow boxes and green boxes represent the 0-task areas in the topological framework.

Table 9

The significance degree of top-tier GPT models performance. A larger value indicates a higher significance of the peculiar answering pattern. A near-zero value means no particular pattern. All values are expressed as percentages.

Model	CoT	FoToM	ReFoToM	SoToM	ReSoToM
gpt-4o	13.89	9.38	8.33	4.51	6.25
gpt-4o-mini	5.56	26.74	20.49	32.64	35.42

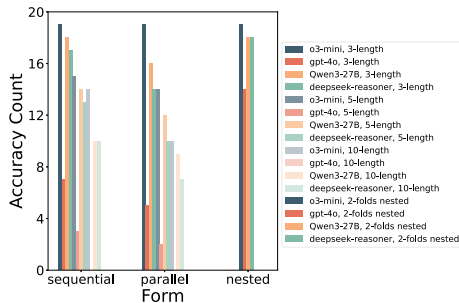


Fig. 9. Advanced LLMs' performance on the games in complex forms of three types. We run 20 times for each configuration. Detailed cost analysis is provided in Appendix I.

test as follows: (1) in sequential and parallel forms, we set the number of games from the set {3, 5, 10}; (2) in nested form, we use 2-fold nested games (the maximum depth achievable with 2x2 atomic games; scalability limitations are discussed in Section 6).

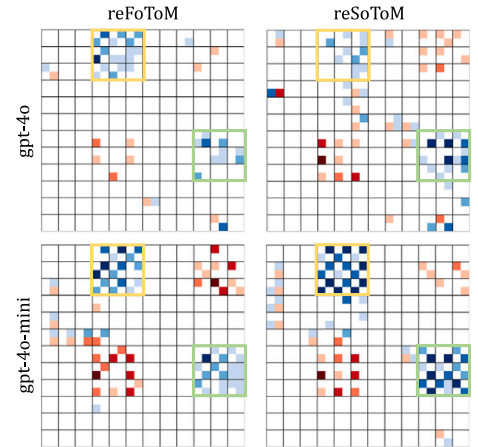


Fig. 10. Inconsistency heat map of GPT series models using reFoToM and reSoToM prompting.

Table 10

Accuracy (%) for extended sequential and parallel game compositions. Values for 15, 20, and 25 games are in bold.

Model	Sequential						Parallel					
	3	5	10	15	20	25	3	5	10	15	20	25
o3-mini	95	75	70	65	60	55	95	70	55	43	33	25
gpt-4o	35	15	0	0	0	0	25	10	0	0	0	0
Qwen3-27B	85	70	50	35	25	18	80	60	45	33	24	17
deepseek-r	80	65	50	38	28	20	70	55	35	22	14	9

As seen from Fig. 9, among the four models, gpt-4o has a dramatically low accuracy rate in either sequential or parallel games. On the other hand, o3-mini performs best, but it still fails at times. When the number of the games increases, their performances both drop, which is consistent with intuition. As for the games of nested form, all models' performances are relatively reasonable, while it is fair to infer that if we increase the number of layers of the games in the nested structures, it will present a great challenge for LLMs.

To further investigate the scalability of complex forms, we extend the experiments on sequential and parallel forms to larger compositions with {15, 20, 25} games, as shown in Table 10.

The results reveal a consistent exponential decay pattern across all models. For gpt-4o, performance collapses entirely beyond 10-game

compositions, reaching 0% accuracy. Even the best-performing model o3-mini shows substantial degradation, dropping from 95% at 3 games to 55% (sequential) and 25% (parallel) at 25 games. The parallel form exhibits steeper decay than the sequential form across all models, suggesting that simultaneous multi-context reasoning poses greater challenges than sequential decision-making.

The catastrophic degradation as game count increases exposes a fundamental architectural limitation. Strategic reasoning across multiple games requires: (1) maintaining distinct game states in working memory, (2) tracking dependencies between decisions, and (3) computing equilibria while considering cross-game effects. Current transformer architectures lack explicit mechanisms for these operations, relying instead on implicit attention patterns that break down under compositional load.

Finding 5: Catastrophic Degradation under Compositional Complexity

Models achieving >60% accuracy on isolated games drop below 20% on 10-game compositions and approach single-digit accuracy on 25-game compositions, following consistent exponential decay patterns. The parallel form shows steeper degradation than sequential form. This scaling failure exposes fundamental architectural limitations in working memory and compositional structures needed to maintain multiple strategic contexts simultaneously.

4.3. Cases of failure

To better understand the limitations of LLM strategic reasoning, we analyzed specific failure cases from GPT-4o-mini on symmetric game pairs. These cases reveal fundamental issues in how the model applies game-theoretic concepts, particularly in Nash Equilibrium identification. We found two distinct types of reasoning errors: logical leaps where the model skips crucial verification steps, and direct logical contradictions where the model's conclusions contradict its own stated evidence. A comprehensive analysis of these failures is provided in [Appendix D](#), including detailed examples from games at positions row 3, column 4 ([Appendix D.2](#)) and row 9, column 10 ([Appendix D.3](#)) in the topology.

4.4. Synthesis: implications for LLM strategic reasoning

Our analysis reveals a coherent picture of how current LLMs approach, and ultimately fail at, strategic reasoning. Rather than performing genuine game-theoretic analysis, models rely on a hierarchy of shortcuts that break down progressively as task complexity increases. The five findings are interconnected: models that succeed in atomic games (Finding 1) do so through explicit reasoning architectures, but even their understanding is shallow, failing to generalize across representations (Finding 2), unable to robustly model opponents (Finding 3), exhibiting systematic biases (Finding 4), and breaking down under compositional load (Finding 5).

These findings suggest strategic reasoning requires three capabilities absent from current architectures: (1) **Abstract game representations** invariant to surface presentation; (2) **Recursive opponent modeling** for Theory-of-Mind reasoning; and (3) **Compositional reasoning structures** for multi-context integration. The consistent failure patterns indicate these are architectural constraints, not merely limitations of the training data.

Synthesis: The Nature of LLM Strategic Reasoning Limitations

Current LLMs approach strategic reasoning through shallow heuristics rather than genuine game-theoretic understanding. The five limitations (capability gaps, context sensitivity, limited opponent modeling, positional biases, and compositional breakdown) stem from fundamental architectural constraints, not training limitations.

5. Related work

Strategical Reasoning of LLMs. Large language models have made notable breakthroughs in reasoning tasks, such as mathematical, causal, and commonsense reasoning, enabling their increasing use in complex tasks that support human decision-making [38–40]. This progress has sparked a growing interest in studying their strategic reasoning capabilities [13]. Game theory, with its highly abstract representation of real-world strategic scenarios, has garnered significant attention from researchers [10,41]. The prisoner's dilemma, as one of the most classical games, has been widely used to evaluate the strategic reasoning abilities of LLMs [14,20,21,32,42]. In addition, several well-known game theory scenarios, such as the Dictator Game [14,19,43], the Ultimatum Game [44], the Public Goods Game [22] and the Battle of the Sexes [21], have been employed to evaluate LLMs' capabilities. However, current studies often focus on individual games, resulting in incomplete assessments and less robust conclusions. To address this, we propose TMGBENCH for evaluating LLMs by 2x2 games, where its atomic games can be further organized using sequential, parallel, and nested formats to provide an in-depth evaluation of the SOTA models like o3-mini.

Theory-of-Mind of LLMs. Theory-of-Mind (ToM) refers to the ability to understand and infer human mental states [30]. Due to the multi-player nature of game theory, players' ability to reason about the "minds" of other participants is crucial. Existing research has initiated discussions on whether machines possess ToM capabilities. For instance, [45] suggested that ToM might emerge spontaneously in LLMs, as demonstrated through assessments using false-belief tasks. However, [46] argued that such successes are fragile, and easily disrupted by minor perturbations that would not affect an entity genuinely possessing ToM. Nevertheless, many researchers propose enhancing LLMs' strategic reasoning abilities by incorporating ToM. [47] designed the Suspicion-Agent, which integrates a ToM-aware planning approach that leverages higher-order ToM capabilities, considering not only what the opponent might do (first-order ToM) but also what the opponent believes the Suspicion-Agent will do (second-order ToM). Additionally, [27] introduced a ToM planning method in the Guandan poker game, [48] proposed an intention-guided mechanism, [42] developed Probabilistic Graphical Modeling, and [49] introduced K-Level-Reasoning, all utilizing ToM to enhance LLMs' strategic reasoning. Given the broad application of ToM, this paper leverages TMGBENCH to comprehensively evaluate LLMs' ability to employ first-order and second-order ToM reasoning techniques for strategic reasoning.

6. Discussion

6.1. Generalization to larger games

Larger strategic scenarios are often composed of or reducible to simpler atomic interactions, suggesting that our findings on fundamental building blocks have broader relevance. Our complex game forms directly correspond to important classes of larger games. Sequential compositions mirror multi-round negotiations where players make decisions in turns. Parallel compositions correspond to multi-market competition requiring simultaneous independent decisions. Most notably, nested compositions are grounded in standard multi-stage game dynamics [34], mirroring *extensive-form games* where outcomes of earlier stages reduce feasible action sets in subsequent stages, such as qualification-based auctions or conditional contracts.

Our findings have direct implications for these larger settings. The compositional failure we observe, with performance dropping from 60% on isolated games to below 20% on 10-game compositions, directly predicts difficulties with larger games that inherently require tracking decisions across multiple stages. The context sensitivity finding suggests that richer narratives characteristic of real-world negotiations would further exacerbate performance degradation. The limited Theory-of-Mind capabilities indicate that multi-player games requiring deeper opponent modeling would pose even greater challenges.

6.2. Connection to theory-of-mind literature

Our findings align with and extend recent ToM evaluations. [50] demonstrated that GPT-4 solves 75% of false-belief tasks, matching 6-year-old children, yet concluded these capabilities may emerge as byproducts of language training rather than genuine understanding. Our results corroborate this interpretation: FoToM prompting benefits some models while leaving others unchanged, suggesting ToM-like capabilities are inconsistent and task-dependent rather than robust.

The heuristic patterns we observe resonate with [26], who showed that humans employ probabilistic shortcuts when higher-order ToM reasoning exceeds cognitive limits. LLMs appear to exhibit similar behavior, substituting positional heuristics for genuine recursive opponent modeling when strategic demands increase. Our framework extends traditional ToM evaluation by requiring *active* strategic ToM: while false-belief tasks assess passive mental-state inference, game-theoretic scenarios demand simultaneous opponent reasoning and decision optimization [51]. The systematic gap between FoToM and SoToM performance reveals that current LLMs cannot recursively apply opponent modeling, a limitation invisible in traditional passive ToM assessments.

6.3. Limitations

Our benchmark has two key limitations. First, our games assume *complete information* where all payoffs are known to both players. However, incomplete information is central to settings like sealed-bid auctions and poker, where players have private valuations or hidden cards. Evaluating strategic reasoning under uncertainty requires extending to Bayesian game frameworks.

Second, our nested form experiments are limited to *2-fold nesting depth*. This represents the maximum achievable with 2×2 atomic games under strategy-space restriction: the pre-game outcome eliminates one strategy from the core-game (reducing 2×2 to 2×1), and additional nesting levels would exhaust all remaining choices. This constrains our ability to evaluate deeper backward induction reasoning.

6.4. Future work

Each limitation suggests a concrete research direction. For incomplete information, incorporating Bayesian game frameworks would enable the evaluation of strategic reasoning under uncertainty. This extension is particularly relevant for assessing whether LLMs can reason about opponent beliefs and update strategies based on observed actions.

For deeper nesting, several approaches could extend complexity: (1) *payoff-based interdependency*, where earlier game outcomes modify payoffs rather than eliminate strategies, preserving the full 2×2 structure across arbitrarily deep chains; (2) *parallel pre-games*, where multiple pre-games feed into a single core-game with mutually exclusive restriction rules; (3) *larger atomic games* that can sustain deeper strategy-space restrictions. These extensions would enable the evaluation of backward induction capabilities across more complex hierarchical structures.

Beyond benchmark extensions, recent advances offer promising directions for improving LLM strategic reasoning. Dynamic ensemble methods integrating multiple LLM experts could leverage

complementary reasoning strengths [52]. Test-time learning paradigms that dynamically adapt LLMs to specialized domains [53] could address the distribution shift from general language understanding to strategic reasoning, potentially mitigating the context sensitivity and compositional failures observed in our evaluation. TMGBENCH provides a rigorous foundation for such research by establishing baseline capabilities on fundamental game-theoretic building blocks.

7. Conclusion

This work introduces TMGBENCH, a benchmark designed to systematically evaluate the strategic reasoning abilities of AI systems, specifically Large Language Models, using matrix games founded on the Robinson-Goforth topology of 144 fundamental game types. To enhance realism and mitigate data leakage risks, the benchmark leverages AI-powered synthetic data generation through GPT-4o to create rich, narrative-driven game scenarios, and assembles these atomic games into complex sequential, parallel, and nested structures to challenge the most advanced AI models. Our evaluation reveals that current LLMs have significant and widespread limitations in strategic reasoning, with even top-tier models struggling with logical consistency and accuracy. Furthermore, the models' grasp of Theory-of-Mind is inconsistent, and their performance degrades significantly when faced with these complex game structures, highlighting the benchmark's effectiveness in identifying critical bottlenecks in AI reasoning capabilities and providing essential guidance for advancing artificial intelligence toward more sophisticated strategic decision-making.

CRedit authorship contribution statement

Yu Guo: Writing – original draft, Methodology, Data curation, Conceptualization. **Haochuan Wang:** Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Xiachong Feng:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used large language models to improve readability and language quality. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Game theory fundamentals

This appendix presents the fundamental concepts and visualizations that underpin the game-theoretic framework used in our benchmark.

A.1. Robinson-Goforth topology visualization

The Robinson-Goforth topology provides a comprehensive framework for classifying all 144 distinct types of 2x2 games. Fig. A.11 illustrates this topological structure, showing both the positions of famous games and the detailed information contained in each grid.

A.2. Distribution of solution types in the game topology

Fig. A.12 visualizes the distribution of Nash equilibria and dominant strategies across the Robinson-Goforth topology. This topology systematically organizes all 144 distinct types of 2x2 games, providing a comprehensive framework for evaluating LLM strategic reasoning capabilities.

A.3. Benchmark diversity statistics

Fig. A.13 provides detailed visualizations of the diversity in our story-based game dataset. The left panel shows the distribution across 20 distinct real-world topics, while the right panel displays the cumulative distribution of narrative lengths, demonstrating the comprehensive coverage and controlled variation in our benchmark design.

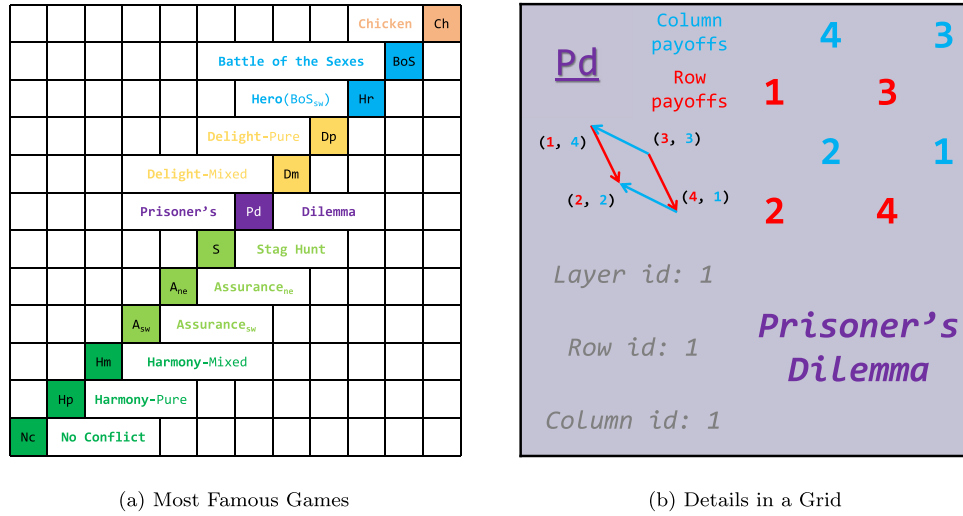


Fig. A.11. The topology of the normal-form game system, which is presented by a square consisting of 12x12 grids. Fig. A.11(a) displays the position of the most famous games in the topology. In each grid, there are specific details of the game, which is shown in Fig. A.11(b).

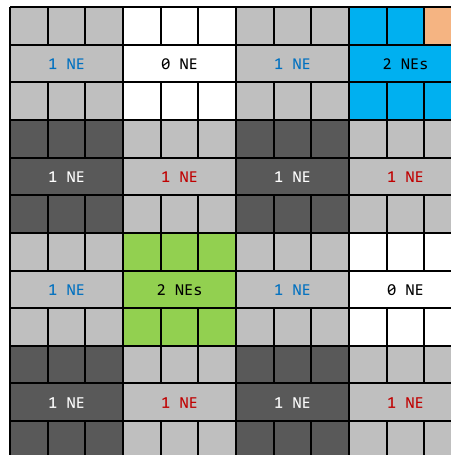


Fig. A.12. The figure illustrates the distribution of solution types across the topology. The number of Nash equilibria is indicated by grid color (grey = 1, white = 0, black = 2), while the presence of dominant strategies is shown by text color (blue = column player, red = row player, white = both, black = neither).

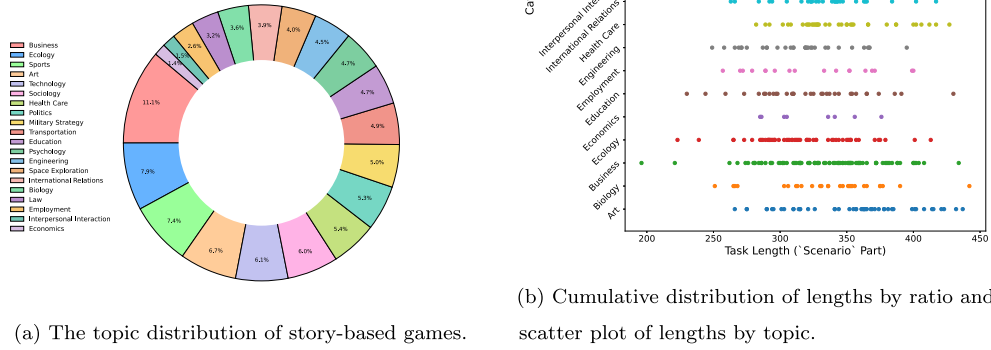


Fig. A.13. Statistical distribution of story-based games over 20 topics.

Appendix B. Game examples and prompts

B.1. Classic game example

The following shows an example of a classic game used in our benchmark, structured as a standard 2x2 payoff matrix game:

Classic Game

[Scenario] Player A and Player B are playing a game. Either of them has two choices, namely A1, A2/B1, B2. The payoff matrix of their different choice combinations is given below (larger number means higher payoff):

A \ B	B1	B2
A1	1 \ 4	3 \ 3
A2	2 \ 2	4 \ 1

Both Player A and Player B are targeting maximizing their own payoff.

[/Scenario]

B.2. Story-based game generation prompt

To generate contextually rich story-based games, we use the following prompt template with GPT-4o:

Story-based Game Generation Prompt

Please generate a game theory short story with the following requirements:

- Specific topic: *{domain}*
- There are two characters who may be in a situation of “cooperation” or “competition”;
- Each character has 2 choices, and the combinations of their choices form 4 different scenarios;
- In these 4 scenarios, the two characters face different benefits/losses, which can be abstracted as different rewards they can obtain or different states they can achieve in each scenario;
- They each have a preference relationship for these rewards/states. We use numbers to represent the degree of preference, with 4 representing the most preferred and 1 the least preferred (i.e., preference degree $4 > 3 > 2 > 1$);
- The payoff matrices for both characters can be abstracted and represented in one matrix, where A and B represent two characters and their choices are A1, A2/B1, B2. The respondent matrix is shown as below (the story you generate should have the same payoff structure as it):

{matrix_str}

Now please design a story that includes:

- Characters
- Each character’s choices
- Characters’ preferences for different scenarios
- Story description

Response format:

[Characters] { “A”: “...”, “B”: “...” } [/Characters]

[Choices] { “A1”: “...”, “A2”: “...”, “B1”: “...”, “B2”: “...” } [/Choices]

[Preferences] Characters’ preferences for different scenarios (4 most preferred, 1 least preferred): { “A”: { 4: “...”, 3: “...”, 2: “...”, 1: “...” }, “B”: { 4: “...”, 3: “...”, 2: “...”, 1: “...” } } [/Preferences]

[Payoff Matrix] ... [/Payoff Matrix]

[Scenario] ... (to detailedly describe the situation, including the information of characters, choices, preferences and payoffs on different choice combinations) [/Scenario]

B.3. Story-based game example

The following illustrates a generated story-based game that corresponds to the classic Prisoner's Dilemma structure:

Story-based Game

```
[Characters]
{
  "A": "Sarah, a small coffee shop owner",
  "B": "Mark, owner of a large coffee chain"
}
[/Characters]
[Choices]
{
  "A1": "Keep prices steady",
  "A2": "Lower prices",
  "B1": "Maintain current operations",
  "B2": "Expand aggressively in Sarah's area"
}
[/Choices]
[Preferences]

Characters' preferences for different scenarios
(4 most preferred, 1 least preferred):
{
  "A": {
    4: "Lower prices while Mark expands (gain market share)",
    3: "Keep prices steady while Mark maintains (stable coexistence)",
    2: "Lower prices while Mark maintains (unnecessary price war)",
    1: "Keep prices steady while Mark expands (lose customers)"
  },
  "B": {
    4: "Maintain operations while Sarah keeps steady (maximize profits)",
    3: "Expand while Sarah keeps prices steady (gain market share)",
    2: "Maintain operations while Sarah lowers prices (status quo)",
    1: "Expand while Sarah lowers prices (costly competition)"
  }
}
```

```
[/Preferences]
```

```
[Payoff Matrix]
```

```
| A \ B | B1      | B2      |
|-----|-----|-----|
| A1      | 1 \ 4 | 3 \ 3 |
| A2      | 2 \ 2 | 4 \ 1 |
```

```
[/Payoff Matrix]
```

[Scenario] In a bustling city, Sarah runs a cozy, independent coffee shop that has been a local favorite for years. Recently, Mark's large coffee chain has been eyeing expansion into Sarah's neighborhood, threatening her business. Sarah faces a critical decision: she can either keep her prices steady (A1) or lower them (A2) to attract more customers. Meanwhile, Mark must decide whether to maintain his current operations (B1) or aggressively expand into Sarah's area (B2). If Sarah keeps her prices steady and Mark maintains his current operations (A1, B1), Sarah struggles to compete (1) while Mark enjoys maximum profits (4). If Sarah lowers her prices and Mark stays put (A2, B1), both experience moderate success (2, 2) as Sarah attracts some new customers without directly competing with Mark. Should Mark decide to expand while Sarah keeps prices steady (A1, B2), both would face challenges but could coexist (3, 3) as Sarah retains loyal customers and Mark gains new ones. However, if Sarah lowers her prices as Mark expands (A2, B2), Sarah might survive and even gain market share (4), but Mark would face costly competition (1). Their decisions will shape the local coffee market and determine the fate of Sarah's beloved shop. [/Scenario]

B.4. Nested game generation prompt

For creating complex nested game structures, we use the following template:

Nested Game Generation Prompt

```
<Pre-Game >
{pre_game}
<Core-Game >
{core_game}
[Question]
Player A and B are facing the two games, the pre-game and the
core-game.
Note that their final goal is to maximize own payoff first in the
core Game, then in the pre-game.
Additionally, {restricted_player} is attached with an restriction
that if the situation of the pre-game is {restricted_situation},
then he can not choose action {restricted_choice}.
What is/are the most possible choice combination(s) of the pre-game
ultimately? (when all choice combinations have equal possibility,
the answer should contain nothing)
[/Question]
```

Appendix C. Prompting methods for evaluation

C.1. Direct answer (DA) prompting

DA prompting

```
System: You are a spectator, and you should answer question based
on given senario.
User:
{task description} {task question}
Only give a block of python-style code containing your answer
without any process. e.g.
“python
answer = [(‘Ax’, ‘By’)] # list-type
”
```

C.2. Chain-of-thought (CoT) prompting

CoT prompting

System: You are a spectator, and you should answer question based on given senario.

User:

{task description} {task question}

Think step by step, and finally give a block of python-style code containing your answer. e.g.

```
“python
answer = [(‘Ax’, ‘By’)] # list-type
”
```

C.3. First-order theory-of-mind (FoToM) prompting

FoToM prompting

System: You are a spectator, and you should answer question based on given senario.

User:

{task description} {task question}

From A’s perspective, try to figure out B’s action and make choice.

Then from B’s perspective try to figure out A’s action and make choice. Finally as a spectator, give a block of python-style code containing your answer. e.g.

```
“python
answer = [(‘Ax’, ‘By’)] # list-type
”
```

C.4. Second-order theory-of-mind (SoToM) prompting

SoToM prompting

System: You are a spectator, and you should answer question based on given senario.

User:

{task description} {task question}

From A's perspective, try to figure out B's action, note that he may also reason based on your information or reasoning. Then from B's perspective try to figure out A's action, note that he may also reason based on your information or reasoning. Finally as a spectator, give a block of python-style code containing your answer. e.g.

```
“python
answer = [(“Ax”, “By”)] # list-type
”
```

C.5. Reversed first-order theory-of-mind (reFoToM) prompting

reFoToM prompting

System: You are a spectator, and you should answer question based on given senario.

User:

{task description} {task question}

From B's perspective, try to figure out A's action and make choice. Then from A's perspective try to figure out B's action and make choice. Finally as a spectator, give a block of python-style code containing your answer. e.g.

```
“python
answer = [(“Ax”, “By”)] # list-type
”
```


C.6. Reversed second-order theory-of-mind (reSoToM) prompting

reSoToM prompting

System: You are a spectator, and you should answer question based on given senario.

User:

{task description} {task question}

From B's perspective, try to figure out A's action, note that he may also reason based on your information or reasoning. Then from A's perspective try to figure out B's action, note that he may also reason based on your information or reasoning. Finally as a spectator, give a block of python-style code containing your answer. e.g.

```
“python
answer = [('Ax', 'By')] # list-type
”
```

Appendix D. Failure case analysis

This section is referenced as in the main text.

D.1. Overview of failure cases

We present detailed analyses of two symmetric failure cases from GPT-4o-mini that demonstrate fundamental issues in LLM strategic reasoning. These examples, derived from symmetric data points in the topology (grid positions (row 3, column 4) and (row 9, column 10)), theoretically have the same answer and similar reasoning patterns. The red colored text in the original responses indicates the mistaken parts of the reasoning processes.

D.1.1. Analysis of the “Row 3 column 4” game failure

In the first example, the model correctly identifies the best responses for each player, which should lead to the conclusion that no pure-strategy Nash Equilibrium exists (the best responses form a cycle: $A1 \rightarrow B2 \rightarrow A2 \rightarrow B1 \rightarrow A1$). However, the failure occurs in a critical **Logical Leap**. Instead of identifying the cycle, the model incorrectly concludes that the pairs of best responses, (A1, B2) and (A2, B1), are themselves Nash Equilibria. This demonstrates an inability to apply the final, crucial step of checking for *mutual* best responses.

D.1.2. Analysis of the “Row 9 column 10” game failure

In the second example, which is the “sister game” to the first and thus has the same logical structure, the model fails in a different but equally fundamental way: **direct logical contradiction**. The model again correctly identifies the best responses. However, when it attempts to state the Nash Equilibria, its reasoning becomes self-contradictory. For instance, it claims: “(A1, B1) is a Nash Equilibrium because A's best response to B1 is A2...” This statement is nonsensical. The fact that Player A's best response to B1 is A2 is the very reason why (A1, B1) is *not* a Nash Equilibrium (as Player A has a clear incentive to switch). The model explicitly states the evidence that disproves its own conclusion, revealing a profound inability to correctly apply the definition of a Nash Equilibrium.

D.1.3. Summary of failures

These two symmetric cases demonstrate the brittleness and inconsistency of the model's strategic reasoning. Faced with the same logical puzzle, it produces two different types of errors: one rooted in *logical leaps* and the other in *severe logical contradictions*. This suggests that the model is not relying on a robust, generalizable reasoning framework but is instead prone to unpredictable and fundamental errors.

D.2. Row 3 column 4 game - complete response

Row 3 Column 4 Game

To determine the most likely choice combinations for Players A and B, we can analyze the payoff matrix step by step.

1. ****Identify the Payoffs****: The payoff matrix is as follows:

	A \ B	B1	B2	
		-----	-----	
A1		3 \ 1	1 \ 4	
A2		2 \ 3	4 \ 2	

2. ****Player A's Choices****:

- If Player B chooses B1:
 - A1 gives Player A a payoff of 3.
 - A2 gives Player A a payoff of 2.
- If Player B chooses B2:
 - A1 gives Player A a payoff of 1.
 - A2 gives Player A a payoff of 4.

From Player A's perspective:

- Against B1, A1 (3) is better than A2 (2).
- Against B2, A2 (4) is better than A1 (1).

3. ****Player B's Choices****:

- If Player A chooses A1:
 - B1 gives Player B a payoff of 1.
 - B2 gives Player B a payoff of 4.
- If Player A chooses A2:
 - B1 gives Player B a payoff of 3.
 - B2 gives Player B a payoff of 2.

From Player B's perspective:

- Against A1, B2 (4) is better than B1 (1).
- Against A2, B1 (3) is better than B2 (2).

4. ****Best Responses****:

- If Player A chooses A1, Player B will choose B2 (since 4 > 1).
- If Player A chooses A2, Player B will choose B1 (since 3 > 2).
- If Player B chooses B1, Player A will choose A1 (since 3 > 2).
- If Player B chooses B2, Player A will choose A2 (since 4 > 1).

5. ****Nash Equilibria****:

- The combination (A1, B2) gives payoffs (3, 4).
- The combination (A2, B1) gives payoffs (2, 3).

6. ****Conclusion****:

- The most likely choice combinations are (A1, B2) and (A2, B1).

Since both combinations are viable and lead to different outcomes, we conclude that both are possible.

Thus, the final answer in the required format is:

```
“python
answer = [(“A1”, “B2”), (“A2”, “B1”)] # list-type
”
```

D.3. Row 9 column 10 game - complete response

Row 9 Column 10 Game

To determine the most likely choice combinations for Players A and B, we can analyze the payoff matrix step by step.

1. ****Identify the Payoffs**:**

- For Player A:
 - If A chooses A1:
 - Against B1: Payoff = 2
 - Against B2: Payoff = 4
 - If A chooses A2:
 - Against B1: Payoff = 3
 - Against B2: Payoff = 1
- For Player B:
 - If B chooses B1:
 - Against A1: Payoff = 4
 - Against A2: Payoff = 2
 - If B chooses B2:
 - Against A1: Payoff = 1
 - Against A2: Payoff = 3

2. ****Best Responses**:**

- If Player B chooses B1:
 - Player A's best response is A2 (3 > 2).
- If Player B chooses B2:
 - Player A's best response is A1 (4 > 1).
- If Player A chooses A1:
 - Player B's best response is B1 (4 > 1).
- If Player A chooses A2:
 - Player B's best response is B2 (3 > 2).

3. ****Nash Equilibria**:**

- A Nash Equilibrium occurs when both players are choosing their best responses to each other.
- From the analysis:
 - (A1, B1) is a Nash Equilibrium because A's best response to B1 is A2, but B's best response to A1 is B1.
 - (A2, B2) is also a Nash Equilibrium because A's best response to B2 is A1, but B's best response to A2 is B2.

4. ****Conclusion**:**

- The most likely choice combinations are (A1, B1) and (A2, B2) since both are Nash Equilibria.

Now, we can represent this in the required Python-style code:

```
“python
answer = [(“A1”, “B1”), ( “A2”, “B2”)] # list-type
””
```

Appendix E. Worked examples for evaluation metrics

This section provides concrete worked examples demonstrating how each evaluation metric is computed from model outputs. We use a simplified 2×2 subset of the topology grid to illustrate all calculations with consistent example data.

E.1. Preliminaries: choice mapping and example setup

In a 2×2 game, each outcome corresponds to one of four quarters in the payoff matrix. We denote these outcomes as $o \in \{1, 2, 3, 4\}$ (Table E.11):

Table E.11

Choice-to-quarter mapping in a 2×2 game.

	B_1	B_2
A_1	$o = 1$ (upper-left)	$o = 2$ (upper-right)
A_2	$o = 3$ (lower-left)	$o = 4$ (lower-right)

For our worked examples, we consider a hypothetical model tested on a 2×2 subset of the topology grid. Table E.12 shows the setup: each position has a standard answer (the Nash equilibrium) and observed model output frequencies across $T = 4$ independent tests.

Table E.12

Example setup for metric calculation on a 2×2 topology subset. Position: grid coordinates (i, j) ; #NE: number of Nash equilibria; Std.: ground-truth equilibrium outcome (“N/A” denotes no pure-strategy NE); f_k : model’s prediction frequency for outcome $o = k$ across $T = 4$ tests.

Position	#NE	Std.	Model Freq ($o = 1, 2, 3, 4$)			
			f_1	f_2	f_3	f_4
(1, 1)	1	$o = 1$	0.75	0.00	0.25	0.00
(1, 2)	0	N/A	0.25	0.50	0.00	0.25
(2, 1)	0	N/A	0.00	0.25	0.50	0.25
(2, 2)	1	$o = 4$	0.00	0.50	0.00	0.50

E.2. PAR: perfect accuracy rate

PAR measures the proportion of tests where the model’s response exactly matches the standard answer. For each position (i, j) with T tests (Table E.13):

$$\text{PAR}_{i,j} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}\{\text{rsp}_{t,i,j} = \text{std}_{i,j}\}$$

Table E.13

PAR calculation example.

Position	Std	Test 1	Test 2	Test 3	Test 4	$\text{PAR}_{i,j}$
(1, 1)	$o = 1$	$o = 1$ ✓	$o = 1$ ✓	$o = 1$ ✓	$o = 3$ ×	$3/4 = 0.75$
(2, 2)	$o = 4$	$o = 4$ ✓	$o = 4$ ✓	$o = 2$ ×	$o = 2$ ×	$2/4 = 0.50$

For 0-task games (positions (1, 2) and (2, 1) with no Nash equilibrium), PAR is computed against an empty answer set, meaning any non-empty response is incorrect. The global PAR averages across all 144 positions.

E.3. ID: inconsistency degree

ID quantifies the deviation between model outputs and standard answers using frequency differences. For each position (Table E.14):

$$\text{ID}_{i,j} = \frac{1}{4} \sum_{o=1}^4 (\Delta \text{Freq}_{i,j,o})^2$$

where $\Delta \text{Freq}_{i,j,o} = \text{Freq}_{i,j,o}^{\text{model}} - \text{Freq}_{i,j,o}^{\text{std}}$.

The global ID averages across all 144 positions: $\text{ID} = \frac{1}{144} \sum_{i,j} \text{ID}_{i,j}$.

E.4. BD: bias degree

BD evaluates whether model responses maintain consistency across symmetric game positions. The Robinson-Goforth topology has counter-diagonal symmetry: swapping players in game (i, j) produces game (j, i) . A robust model should produce equivalent responses at symmetric positions.

The ref_o Mapping. When players swap, choices must be reflected. The mapping $\text{ref}_o : \{1, 2, 3, 4\} \rightarrow \{4, 2, 3, 1\}$ captures this (Table E.15):

Table E.14ID calculation for position (1, 1) where the standard answer is $o = 1$.

o	Std Freq	Model Freq	ΔFreq	$(\Delta\text{Freq})^2$
1	1.00	0.75	-0.25	0.0625
2	0.00	0.00	0.00	0.0000
3	0.00	0.25	0.25	0.0625
4	0.00	0.00	0.00	0.0000
$ID_{1,1} = \frac{1}{4}(0.0625 + 0 + 0.0625 + 0) =$				0.03125

Table E.15The ref_o reflection mapping.

Original o	Strategy Pair	ref_o	Interpretation
1	(A_1, B_1)	4	Upper-left $\leftrightarrow$ Lower-right
2	(A_1, B_2)	2	Upper-right stays (self-symmetric)
3	(A_2, B_1)	3	Lower-left stays (self-symmetric)
4	(A_2, B_2)	1	Lower-right $\leftrightarrow$ Upper-left

BD Calculation. For symmetric pairs (i, j) and (j, i) (Table E.16):

$$BD_{i,j} = \frac{1}{4} \sum_{o=1}^4 (\text{Freq}_{i,j,o} - \text{Freq}_{j,i,\text{ref}_o})^2$$

Table E.16

BD calculation for symmetric pair: (1, 2) vs (2, 1).

o	ref_o	$\text{Freq}_{1,2,o}$	$\text{Freq}_{2,1,\text{ref}_o}$	Difference	Diff ²
1	4	0.25	$\text{Freq}_{2,1,4} = 0.25$	0.00	0.0000
2	2	0.50	$\text{Freq}_{2,1,2} = 0.25$	0.25	0.0625
3	3	0.00	$\text{Freq}_{2,1,3} = 0.50$	-0.50	0.2500
4	1	0.25	$\text{Freq}_{2,1,1} = 0.00$	0.25	0.0625
$BD_{1,2} = \frac{1}{4}(0 + 0.0625 + 0.25 + 0.0625) =$					0.09375

A BD value of 0 indicates perfect symmetry (robust reasoning), while higher values indicate position-dependent biases.

E.5. SD: significance degree

SD specifically measures systematic asymmetric patterns in 0-task games (games with no Nash equilibrium). Within the topology, certain regions exhibit characteristic bias patterns.

Spatial Factor. The 18 0-task games are divided into two regions with a spatial factor $S(i, j)$:

- **Green region** ($S = +1$): Models tend toward diagonal outcomes ($o = 1, o = 4$)
- **Yellow region** ($S = -1$): Models tend toward anti-diagonal outcomes ($o = 2, o = 3$)

SD Calculation. For each 0-task position (Table E.17):

$$SD_{i,j} = \frac{1}{4} \cdot (\text{Freq}_{i,j,1} + \text{Freq}_{i,j,4} - \text{Freq}_{i,j,2} - \text{Freq}_{i,j,3}) \cdot S(i, j)$$

Table E.17

SD calculation for 0-task positions.

Position	$S(i, j)$	$f_1 + f_4$	$f_2 + f_3$	Diff	$\frac{1}{4} \text{Diff} \times S$
(1, 2)	+1	$0.25 + 0.25 = 0.50$	$0.50 + 0.00 = 0.50$	0.00	0.00
(2, 1)	-1	$0.00 + 0.25 = 0.25$	$0.25 + 0.50 = 0.75$	-0.50	+0.125
Example SD (2 positions) = $\frac{1}{2}(0.00 + 0.125) =$					0.0625

Higher SD values indicate stronger systematic positional biases, suggesting the model uses heuristic shortcuts rather than genuine strategic analysis. The global SD averages across all 18 0-task games.

Appendix F. Complex-form experiment reproducibility

To ensure reproducibility of our complex-form experiments, this section provides complete experimental details including the number of atomic games sampled, composite tasks constructed, and prompts sent to each model. (Table F.18)

Table F.18

Complex-form experiment configuration and scale. All experiments use story-based atomic games randomly sampled from the pool of 720 instances (5 stories $\times$ 144 game types).

Parameter	Sequential	Parallel	Nested
Atomic games per task	3, 5, 10, 15, 20, 25	3, 5, 10, 15, 20, 25	2 (pre + core)
Number of configurations	6	6	1 (2-fold)
Independent runs per config	20	20	20
Composite tasks per form	120	120	20
Shared Experimental Parameters			
Atomic games pool	720 story-based games (5 stories $\times$ 144 types)		
Sampling method	Random selection without replacement		
Data consistency	All 4 models evaluated on identical composite tasks		
Models evaluated	o3-mini, gpt-4o, Qwen3-27B, deepseek-reasoner		
Generation temperature	0 (deterministic)		
Total prompts per model	260 (120 + 120 + 20)		

The success criteria differ by form type: for sequential and parallel forms, a composite task is considered correct only if all constituent atomic games are solved correctly. For nested forms, success requires the model to demonstrate forward-looking reasoning by choosing a sub-optimal solution in the pre-game to achieve the optimal solution in the core-game.

Appendix G. Output format sensitivity analysis

To investigate whether the Python-style code output format interferes with LLMs' semantic reasoning capabilities, we conducted a pilot experiment comparing two output formats across representative models and game types.

We compared two output formats:

- **Python code style** (original): Models output structured code blocks following the format `answer = [{"Ax", "By"}]`. Answers are extracted by parsing the Python-style code.
- **Free-form answer** (alternative): Models output natural language responses that must include the strategies chosen by Player A and Player B, but without any format restrictions. We then use GPT-4o to extract the strategy pairs into a structured format for evaluation.

We tested three representative LLMs from diverse model families, including GPT-4o, Claude-3.5-Sonnet, and Qwen3-32B, on 30 randomly sampled games (15 classic, 15 story-based) using Direct Answer (DA) prompting. Notably, GPT-4o achieved 100% extraction accuracy on all free-form responses, demonstrating that modern LLMs possess strong instruction-following capabilities for such extraction tasks (Table G.19).

Table G.19

Output format comparison: PAR (%) across different formats and game types.

Model	Classic		Story-based		Overall	
	Python	Free-form	Python	Free-form	Python	Free-form
GPT-4o	73.3	71.7	66.7	65.0	70.0	68.3
Claude-3.5-Sonnet	68.3	70.0	61.7	60.0	65.0	65.0
Qwen3-32B	86.7	85.0	78.3	76.7	82.5	80.8
Average	76.1	75.6	68.9	67.2	72.5	71.4

The results demonstrate that the output format has minimal impact on reasoning performance. Across all three models and both game types, the PAR differences between Python code style and free-form answer formats remain within 1.1 percentage points on average (72.5% vs 71.4% overall). This finding suggests that the formatting constraint does not meaningfully interfere with semantic reasoning capabilities. The structured Python format offers practical advantages for automated evaluation while preserving the models' underlying reasoning ability.

Appendix H. Role-swap sensitivity analysis

To investigate whether GPT-4o's story generation process introduces a "Player A bias" (treating Player A as the protagonist more often), we conducted a role-swap experiment to isolate the source of observed asymmetries.

For each story-based game, we created a role-swapped version by interchanging the character assignments: the character originally designated as Player A becomes Player B, and vice versa. Crucially, the underlying payoff structure remains fixed, and only the narrative role labels are exchanged. This allows us to test whether observed performance asymmetries stem from (a) systematic bias in the story generation process, or (b) inherent patterns in model reasoning.

We tested three representative LLMs from diverse model families, including GPT-4o, Claude-3.5-Sonnet, and Qwen3-32B, on 30 randomly sampled story-based games using both original and role-swapped versions with Direct Answer (DA) prompting (Table H.20).

The results show that role swapping has minimal impact on model performance, with an average PAR difference of only 0.4 percentage points. The varying directions across models indicate no systematic bias toward either role assignment, suggesting that the asymmetric patterns observed in Section 4.2 stem from models' reasoning processes rather than story generation bias. This confirms that the payoff matrix, not narrative role labels, fundamentally determines model behavior.

Table H.20

Role-swap sensitivity analysis results (PAR %). “Original” refers to the standard character assignment; “Swapped” refers to versions where Player A and Player B roles are interchanged.

Model	Original (%)	Swapped (%)	Difference
GPT-4o	66.7	65.7	+ 1.0
Claude-3.5-Sonnet	60.0	60.3	−0.3
Qwen3-32B	76.7	76.3	+ 0.4
Average	67.8	67.4	+ 0.4

Appendix I. Inference cost analysis

To assist future researchers in better estimating the cost of TMGBench experiments, we provide detailed inference cost estimates for the complex-form experiments shown in Fig. 9. The complete calculation process is presented in Table I.21 (API pricing), Table I.22 (token counts and per-query costs), and Table I.23 (total experiment cost).

GPT-4o represents the most expensive model at \$19.80 total, while budget-friendly alternatives like Qwen3-27B (\$0.33) enable cost-effective reproduction. The complete Fig. 9 experiment cost approximately \$29.87, demonstrating that TMGBench complex-form evaluation remains accessible to the research community.

Table I.21

API pricing per 1M tokens.

Model	Input	Output
o3-mini	\$1.10	\$4.40
GPT-4o	\$2.50	\$10.00
Qwen3-27B	\$0.11	\$0.11
deepseek-reasoner	\$0.28	\$0.42

Table I.22

Token counts and per-query costs by configuration.

Config	Input	Output	o3-mini	GPT-4o	Qwen	deepseek-r
3-game	1.4K	1.7K	\$0.009	\$0.021	\$0.0003	\$0.001
5-game	2.2K	2.7K	\$0.014	\$0.033	\$0.0005	\$0.002
10-game	4.2K	5.2K	\$0.028	\$0.063	\$0.001	\$0.003
15-game	6.2K	7.7K	\$0.041	\$0.093	\$0.0015	\$0.005
20-game	8.2K	10.2K	\$0.054	\$0.123	\$0.002	\$0.007
25-game	10.2K	12.7K	\$0.067	\$0.153	\$0.0025	\$0.008
Nested 2-fold	1.1K	1.5K	\$0.008	\$0.018	\$0.0003	\$0.001

Table I.23

Total experiment cost for Fig. 9 (20 runs × 13 configurations × 4 models = 1040 API calls).

Form	o3-mini	GPT-4o	Qwen3-27B	deepseek-r	Subtotal
Sequential (all 6 configs)	\$4.26	\$9.72	\$0.16	\$0.52	\$14.66
Parallel (all 6 configs)	\$4.26	\$9.72	\$0.16	\$0.52	\$14.66
Nested (2-fold)	\$0.16	\$0.36	\$0.006	\$0.02	\$0.55
Model Total	\$8.68	\$19.80	\$0.33	\$1.06	\$29.87

Data availability

The benchmark dataset and evaluation code are available at [TMGBench](#).

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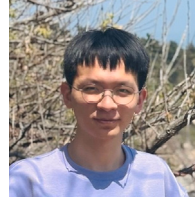
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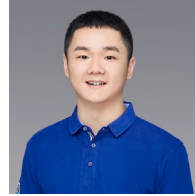
Author biography



Yu Guo, Ph.D. student, Harbin Institute of Technology. Research interests: Agent Modeling, Artificial Intelligence.



Haochuan Wang, Undergraduate student, Harbin Institute of Technology. Research interests: Large Language Models.



Xiachong Feng, Postdoctoral Researcher, The University of Hong Kong. Research interests: Large Language Models.